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CONFORMAL PREDICTION OF TEMPORAL DATA

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Chapter 1

Introduction

This thesis proposes a method for time series uncertainty quantification with conformal prediction. The work addresses two important challenges: constructing a predictive model and proving formal guarantees for its coverage properties under the weakest possible assumptions. The main objective of this thesis is to extend the classical Adaptive Conformal Prediction (Gibbs and Candès, 2021) framework by integrating clustering into its calibration pipeline, while preserving as many of the original theoretical guarantees as possible. This connection is motivated by the practical reality that real-world time series exhibit disturbances and regions of varying model performance. By conditioning the conformal calibration on cluster membership, the method ensures that predictions are informed by the richest available structure in the data rather than aggregated over a homogeneous regime. The theoretical contributions are complemented by an empirical evaluation in which experimental results demonstrate the characteristics of the proposed method and its superiority over classical approaches. Similar methods has wide range of practical applications in industry.

In energy markets reliable prediction intervals are essential for both producers and consumers. The forecast of electricity prices day ahead and balancing markets requires not just an expected price, but a range of values with quantified probability directly used for forming bidding strategies, storage arbitrage and balancing operations (O'Connor et al., 2025).

In manufacturing operations conformal anomaly detection over sensor streams enables predictive maintenance by identifying readings that deflect significantly from standard behavior. Unlike approaches with constant threshold, conformal methods provide a statistically false alarm rate guarantee, reducing maintenance interventions (Yuan et al., 2026).

In logistics, conformal predictive distributions on order fulfillment times enable more reliable delivery promises to customers and more efficient warehouse allocation (Ye et al., 2025).

Chapter 2

Background

2.1 Time Series Conformal Prediction

Conformal prediction is a statistical method designed to provide reliable measures of uncertainty for machine learning models originally presented in Vovk et al., 2005 (reliable introduction in Angelopoulos and Bates, 2022). For any forecasting model which produces single prediction that method allows generate prediction intervals - a range of values expected to contain the true value of the time series with a previously specified at least $(1 - \alpha)$ probability. A typical conformal prediction workflow begins by splitting the available data into several subsets:

- **Training set** – used to train a forecasting model (e.g. Random Forest) that produces point predictions for the time series;
- **Calibration set** – used to compute nonconformity scores - differences between the predictions generated by the trained model and the true observed values of the time series;
- **Test set/Validation set** – used to evaluate the model and generate prediction intervals for unseen observations. The nonconformity scores calculated on the calibration set are used to determine the margin added to the forecasting model point predictions;

Nonconformity scores are typically defined as the absolute residuals $|y_i - \hat{y}_i|$, where y_i denotes the true value and $\hat{y}_i$ denotes the model's point prediction on the calibration set (Angelopoulos and Bates, 2022). Given a desired coverage level $(1 - \alpha)$ the conformal quantile $\hat{Q}_{1-\alpha}$ is computed as the $\lceil (1 - \alpha)(1 + 1/n) \rceil$-th empirical quantile of the collected nonconformity scores, where n is the size of the calibration set. The resulting prediction interval for a new test observation x_t is then constructed as $[\hat{y}_t - \hat{Q}_{1-\alpha},\ \hat{y}_t + \hat{Q}_{1-\alpha}]$ - guaranteeing marginal coverage under the assumption that the data are exchangeable.

One of the key advantages of conformal prediction is that it can be applied with any forecasting algorithm without modifying. The standard exchangeability assumption may be violated in real time series due to temporal dependencies and distributional shifts. Extensions like ACP or SPCI aim to maintain valid coverage even in non-stationary data by updating the nonconformity score distribution when new observations arrive.

2.2 Adaptive Conformal Prediction

Standard Conformal Prediction relies on a fixed level of miscoverage α that remains constant through the evaluation period. It produces prediction intervals of uniform width determined by the calibration set. This approach provides theoretical coverage guaranties only under the assumption of exchangeability.

Let $\gamma \geq 0$ be a step-size parameter and $\alpha \in (0, 1)$ be the target miscoverage level - expected probability that true value is not predicted intervals.

Split the most recent observations indexes into a training set D_{Tr}, calibration set $\mathrm{D}_{\mathrm{Cal}}$ and test set D_{Te}. Using D_{Tr}, fit a prediction model $\hat{\mu}(\cdot)$.



Figure 2.1: Data split diagram.

2.3 Conformal Prediction (CP)

Define the conformity scores on the calibration set as

$$C = \left\{ |Y_i - \hat{\mu}(X_i)| : i \in \mathrm{D}_{\mathrm{Cal}} \right\}.$$

Let $\widehat{Q}_{1-\alpha}(C)$ denote the empirical $(1-\alpha)$-quantile of these scores. The conformal prediction interval at time t is then defined as

$$\widehat{C}_\alpha(X_t) = \left[\hat{\mu}(X_t) \pm \widehat{Q}_{1-\alpha}(\mathrm{C}) \right].$$

2.4 Adaptive Conformal Prediction (ACP)

The method described previously is not resistant to changes in time series distribution. Especially when the conformal set contains only nonconformity scores calculated when the model operated under conditions where it had completely different performance.

A well-known solution to neutralize such behaviour is Adaptive Conformal Prediction Gibbs and Candès, 2021, where the conformal set is of constant length, calculated online by dropping the oldest errors and adding new ones.



Figure 2.2: ACP diagram. R/P represents real and predicted value.

Lets initialize

$$\alpha_1 = \alpha,$$

$$Cal_1 = D_{Cal}.$$

Define the conformity scores on the calibration set as

$$C_{_t} = \left\{ |Y_i - \hat{\mu}(X_i)| : i \in \mathrm{Cal}_t \right\}.$$

Let $\widehat{Q}_{1-\alpha_t}(C_t)$ denote the empirical $(1-\alpha_t)$-quantile of these scores. The adaptive conformal prediction interval at time t in Te is then defined as

$$\widehat{C}_{\alpha_t}(X_t) = \left[\hat{\mu}_t(X_t) \pm \widehat{Q}_{1-\alpha_t}(\mathrm{C}_t)\right].$$

The effective miscoverage level α_t is updated recursively according to

$$\alpha_{t+1} = \alpha_t + \gamma \left(\alpha - \mathbf{1}\{Y_t \notin \widehat{C}_{\alpha_t}(X_t)\}\right).$$

It follows that if in time t real value is covered next interval widens, otherwise narrows.

$$\alpha_{t+1} = \begin{cases} \alpha_t + \gamma(\alpha - 1) \leq \alpha_t & \text{if } Y_t \in \widehat{C}_{\alpha_t}(X_t) \\ \alpha_t + \gamma\alpha \geq \alpha_t & \text{if } Y_t \notin \widehat{C}_{\alpha_t}(X_t) \end{cases},$$

Then drop oldest and add current error to Cal_t (to create updated C for step $t+1$):

$$\mathrm{Cal}_{t+1} = (\mathrm{Cal}_t \cup \{t\}) \setminus \{t - S\}.$$



Figure 2.3: Calibration set sliding in time (t).

This approach has a significant drawback - when the forecasting model encounters noisy data or anomalies large errors dominate the calibration set (D_{Cal}) and have a huge impact on interval lengths. Over the following steps, α and D_{Cal} adapt (speed of recovery depends on S and γ).

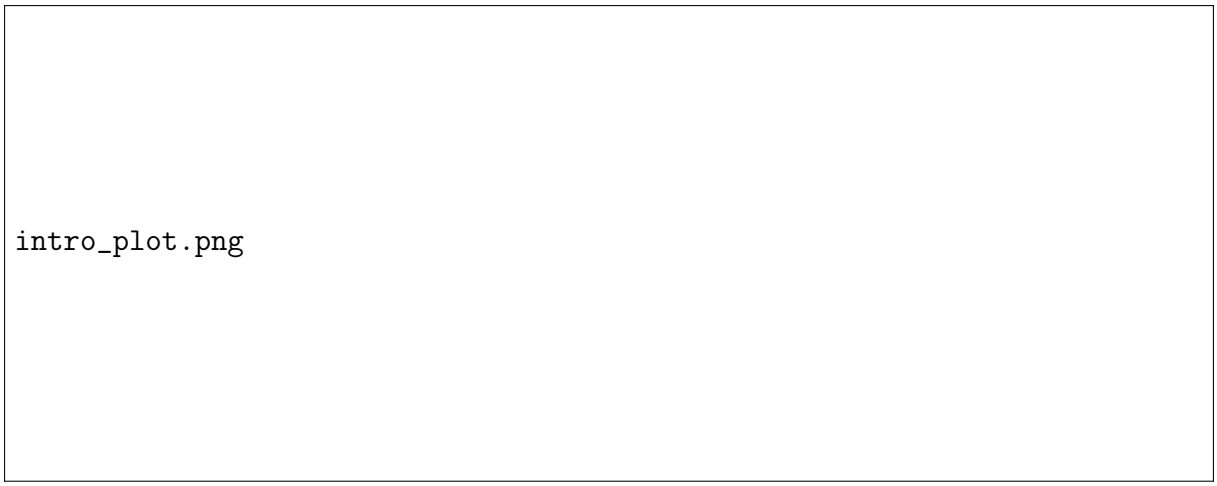


Figure 2.4: Intervals widening example for S = 100 and gamma optimized to reduce average interval length

The proposed clustering-based model was designed to reduce such behavior and react more quickly to time series regime switching.

2.5 Hidden Markov Model

Descriptions were based on "Chapter A Hidden Markov Models" of Jurafsky and Martin., 2026 where more details can be find.

2.5.1 Markov Model

Base for Hidded Markov Model (Leonard E. Baum, 1966) is stochastic Markov Chain. Its definition includes three elements

- **States** – a set $Q = \{q_1, \ldots, q_N\}$ of N distinct elements each representing a possible status of the system.
- **Transition matrix** – an $N \times N$ matrix A, where a_{ij} denotes the probability of moving from state q_i to state q_j or for j = i staying in q_i in a single step. A is invariant, implying the model homogeneity in time.
- **Initial distribution** π – a probability vector specifying the probability that the process $\{Q_t\}$ starts at each state.

The key assumption is the Markov property – the future state depends only on the current state, not on the history of previous states:

$$P(q_{t+1} \mid q_t, q_{t-1}, \ldots, q_1) = P(q_{t+1} \mid q_t).$$

While in a standard Markov Model the states are directly observable, the Hidden Markov Model extends this framework by treating the states as latent variables.

2.5.2 Hidden Markov Model

Assume that we want to calculate probability of events but we cannot see them directly, only their effects are observable. Then classic Markov Model can be redefined as

- **States** – a set $Q = \{q_1, \ldots, q_N\}$ of N distinct elements, each representing a possible status of the system. They are not visible directly.
- **Transition matrix** – an $N \times N$ matrix A, where a_{ij} denotes the probability of moving from state q_i to state q_j or staying in q_i in a single step. A is invariant, implying the model homogeneity in time.
- **Initial distribution** π – a probability vector specifying the probability that the process $\{Q_t\}$ starts at each state.
- **Observations** – a set $V = \{v_1, \ldots, v_M\}$ of M distinct elements, each representing a possible observable effect of the system. Each element could be part of sequence of observation $O = \{o_1 \ldots o_T\}$ where $o_t \in v_1, \ldots v_M$
- **Emission matrix** – an $N \times M$ matrix B, where b_{jk} denotes the probability of observing v_k given that the system is in state $b_{jk} = P(O_t = v_k \mid Q_t = q_j)$.

Apart from Markov property HMM extends assumptions that only latest state has impact on observed effects

$$P(o_{t+1} \mid o_t, \dots o_1, q_{t+1}, \dots q_1) = P(o_{t+1} \mid q_{t+1}).$$

Three main algorithms (Leonard E. Baum, 1970) designed to efficiently use HMM are

- **Evaluation** – for HMM model compute probability of observing given sequence of observations
- **Decoding** – for HMM model compute most likely sequence of hidden states for given sequence of observations
- **Learning** – for given sequence of observations and set of states compute values of transition and emission matrix

Evaluation - Forward Algorithm

Formally evaluation goal is to compute $P(O|\lambda)$ with $\lambda = \{A, B, \pi\}$.

That could be transformed into sum over all N^T possible state sequences:

$$P(O \mid \lambda) = \sum_Q P(O, Q \mid \lambda) = \sum_Q P(O \mid Q, \lambda)\, P(Q \mid \lambda).$$

Where using model assumptions:

$$P(Q \mid \lambda) = \pi_{q_1} \prod_{t=1}^{T-1} a_{q_t q_{t+1}},$$

$$P(O \mid Q, \lambda) = \prod_{t=1}^{T} P(o_t \mid q_t, \lambda) = \prod_{t=1}^{T} b_{q_t o_t}.$$

Computing such formula directly is exponential time, which limits its utility. Luckily storage already computed values using dynamic programming reduce time complexity to $O(N^2T)$.

Key observation for that algorithm acceleration is that

$$P(o_1, \dots, o_t, Q_t = s \mid \lambda) = \sum_{s'=1}^{N} P(o_1, \dots, o_{t-1}, Q_{t-1} = s' \mid \lambda) \cdot a_{s's} \cdot b_s(o_t),$$

and memorizing already computed $P(o_1, \dots, o_{t-1}, Q_{t-1} = s' \mid \lambda)$ during execution.

Decoding

Formally decoding goal is to find the most likely state sequence $Q^* = (q_1, \ldots, q_T)$ given observation sequence O and model λ:

$$Q^* = \arg\max_Q P(Q \mid O, \lambda) = \arg\max_Q P(O, Q \mid \lambda).$$

That could be computed by searching over all N^T possible state sequences, which is exponential time. Again storing already computed values reduces time complexity to $O(N^2T)$ with Viterbi algorithm (Viterbi, 1967).

Key observation for that algorithm acceleration is that

$$\max_{q_1,\ldots,q_{t-1}} P(o_1, \ldots, o_t, Q_t = s \mid \lambda) = \max_{s'} \left[\max_{q_1,\ldots,q_{t-2}} P(o_1, \ldots, o_{t-1}, Q_{t-1} = s' \mid \lambda) \right] \cdot a_{s's} \cdot b_s(o_t),$$

and memorizing already computed $\max_{q_1,\ldots,q_{t-2}} P(o_1, \ldots, o_{t-1}, Q_{t-1} = s' \mid \lambda)$.

The difference from the Forward Algorithm is that instead of summing over all previous states we take the maximum and additionally we store the arg max at each step to reconstruct the optimal path Q^* via backtracking.

Learning - Baum-Welch Algorithm

Formally the learning goal is to find model parameters $\lambda^* = (A, B, \pi)$ that maximize the likelihood of the observation sequence O:

$$\lambda^* = \arg\max_\lambda P(O \mid \lambda).$$

There is no closed form of solution to this optimization problem. The Baum-Welch Algorithm applies Expectation-Maximization to iteratively update parameters until convergence.

The algorithm requires both a forward and a backward pass over the observation sequence. The forward probability $\alpha_t(s)$ is already defined as in the Forward Algorithm. The backward probability is defined analogously as:

$$\beta_t(s) = P(o_{t+1}, \ldots, o_T \mid Q_t = s, \lambda).$$

i.e. the probability of observing the future sequence $o_{t+1}, \ldots, o_T$ given that the system is in state s at time t. It is computed recursively as:

$$\beta_t(s) = \sum_{s'=1}^{N} a_{ss'} \cdot b_{s'}(o_{t+1}) \cdot \beta_{t+1}(s').$$

Combining forward and backward probabilities, we define $\gamma_t(s)$ as the probability of

being in state s at time t given the full observation sequence:

$$\gamma_t(s) = P(Q_t = s \mid O, \lambda) = \frac{\alpha_t(s) \cdot \beta_t(s)}{\sum_{s'=1}^{N} \alpha_t(s') \cdot \beta_t(s')}.$$

Following Bayes theorem - numerator gives the joint probability of the full observation sequence and being in state s at time t, the denominator normalizes distribution.

Similarly, we define $\xi_t(i, j)$ as the probability of transitioning from state i to state j at time t:

$$\xi_t(i, j) = P(Q_t = i, Q_{t+1} = j \mid O, \lambda) = \frac{\alpha_t(i) \cdot a_{ij} \cdot b_j o_{t+1} \cdot \beta_{t+1}(j)}{\sum_{i'=1}^{N} \sum_{j'=1}^{N} \alpha_t(i') \cdot a_{i'j'} \cdot b_{j'} o_{t+1} \cdot \beta_{t+1}(j')}.$$

Using γ_t and ξ_t, the model parameters are re-estimated as:

$$\hat{\pi}_i = \gamma_1(i), \qquad \hat{a}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \qquad \hat{b}_j k = \frac{\sum_{t=1}^{T} \gamma_t(j) \cdot \mathbf{1}[o_t = v_k]}{\sum_{t=1}^{T} \gamma_t(j)}.$$

These re-estimated parameters $\hat{\lambda} = (\hat{A}, \hat{B}, \hat{\pi})$ are then used as the new model in the next iteration, repeating until convergence.

Chapter 3

Proposed method

Let h denote the clustering model sequence length and τ the probability threshold.
Let z_t denote the cluster at time t and let $\mathcal{C}_t^{(k)}$ denote the state-specific conformal set for state k with limited $|\mathcal{C}_t^{(k)}|$ to N.



Figure 3.1: ACP - clustering-based diagram with HMM as clustering model. R and P represents real and predicted value.

The main idea behind that model is to use more conformal sets and choose the active one based on forecasting model performance. Updating $\mathcal{C}^{(k)}$ is done by adding a pair of error value and probability of being at state k at time t, but only for states where this probability exceeds a threshold τ, ensuring that low-confidence states do not accumulate noisy observations. At prediction time the clustering model identifies the most likely current state and only the corresponding conformal set is used to compute the prediction interval.
The quantile is computed as a weighted empirical quantile, where weights reflect how strongly each past observation was associated with the active state. This way the resulting interval adapts not only to the current miscoverage level, but also to the regime identified

by clustering model.

Initialize over all states:

$$\alpha_1 = \alpha.$$

3.1 Calibration Sets Update

For each $i \in \mathrm{Cal}_t$, define the conformity score

$$S_t = |Y_t - \hat{\mu}(X_t)|.$$

Compute posterior state probabilities:

$$w_t^{(k)} = \mathbb{P}\big(z_t = k \mid Y_{t-h+1}, \ldots, Y_t\big).$$

For states satisfying

$$w_t^{(k)} \geq \tau,$$

update the state-specific conformal set:

$$\mathcal{C}_t^{(k)} \leftarrow \mathcal{C}_t^{(k)} \cup \{(S_t, w_t^{(k)})\}.$$

If $|\mathcal{C}_t^{(k)}| > N$ drop oldest element.

3.2 Online Prediction

At time t, compute the point prediction

$$\hat{\mu}(X_t).$$

Estimate the hidden state:

$$\hat{z}_t = \arg\max_k \mathbb{P}\big(z_t = k \mid Y_{t-h+1:t}\big).$$

Let

$$\mathcal{C}_t^{(\hat{z}_t)} = \{(S_1, w_1), \ldots, (S_n, w_n)\}.$$

Normalize weights:

$$\tilde{w}_i = \frac{w_i}{\sum_{j=1}^n w_j}.$$

Sort increasingly:

$$S_{(1)} \leq \cdots \leq S_{(n)}.$$

Define cumulative weights:

$$F_k = \sum_{i=1}^{k} \tilde{w}_{(i)}.$$

Let

$$k_{\min} = \min \{k : F_k \geq 1 - \alpha_t\}.$$

Define the empirical weighted quantile

$$\widehat{Q}_{1-\alpha_t} = S_{(k_{\min})}.$$

The adaptive conformal prediction interval at time t is

$$\widehat{C}_{\alpha_t}(X_t) = \left[\hat{\mu}(X_t) \pm \widehat{Q}_{1-\alpha_t}\right].$$

The miscoverage level is updated recursively:

$$\alpha_{t+1} = \alpha_t + \gamma \left(\alpha - \mathbf{1}\{Y_t \notin \widehat{C}_{\alpha_t}(X_t)\}\right).$$

Using those procedures whole algorithm execution can be presented as 2 parts Train and Calibration phase (models training and calibration sets filling) and Online prediction phase (intervals prediction and online adaptation).

Algorithm 1 Clustering-based ACP – Train and Calibration phase

//Train phase

Train $\hat{\mu}(\cdot)$ and clustering model (ex. HMM) on D_{Tr}.

$\alpha_1 \leftarrow \alpha$

for $k = 1, \ldots, K$ **do**

  $\mathcal{C}_1^{(k)} \leftarrow \emptyset$

end for

//Calibration phase

for $i = 1, \ldots, |\mathcal{D}_{\text{Cal}}|$ **do**

  $S_i \leftarrow |Y_i - \hat{\mu}(X_i)|$

  for $k = 1, \ldots, K$ **do**

    $w_i^{(k)} \leftarrow P(z_i = k \mid Y_{i-h+1}, \ldots, Y_i)$

    if $w_i^{(k)} \geq \tau$ **then**

      $\mathcal{C}^{(k)} \leftarrow \mathcal{C}^{(k)} \cup \left\{ \left(S_i,\, w_i^{(k)} \right) \right\}$

      if $\left| \mathcal{C}^{(k)} \right| > N$ **then**

        Drop oldest element from $\mathcal{C}^{(k)}$

      end if

    end if

  end for

end for

Algorithm 2 Clustering-based ACP – Online prediction phase

for $t = 1, \ldots, |\mathcal{D}_{\text{Te}}|$ **do**

// Identify active state

$\hat{z}_t \leftarrow \arg\max_k \; P(z_t = k \mid Y_{t-h+1}, \ldots, Y_{t-1})$

Let $\mathcal{C}_t^{(\hat{z}_t)} = \{(S_1, w_1), \ldots, (S_n, w_n)\}$

for $i = 1, \ldots, n$ **do**

$\tilde{w}_i \leftarrow w_i \,/\, \sum_{j=1}^{n} w_j$

end for

Sort so that $S_{(1)} \leq \cdots \leq S_{(n)}$

for $k = 1, \ldots, n$ **do**

$F_k \leftarrow \sum_{i=1}^{k} \tilde{w}_{(i)}$

end for

$k_{\min} \leftarrow \min\{k : F_k \geq 1 - \alpha_t\}$

$\hat{Q}_{1-\alpha_t} \leftarrow S_{(k_{\min})}$

$\hat{C}_{\alpha_t}(X_t) \leftarrow \left[\hat{\mu}(X_t) \pm \hat{Q}_{1-\alpha_t}\right]$

// Observe true value and update

Observe Y_t

$\text{err}_t \leftarrow \mathbf{1}\left\{Y_t \notin \hat{C}_{\alpha_t}(X_t)\right\}$

$\alpha_{t+1} \leftarrow \alpha_t + \gamma\,(\alpha - \text{err}_t)$

// Update calibration sets

$S_t \leftarrow |Y_t - \hat{\mu}(X_t)|$

for $k = 1, \ldots, K$ **do**

$w_t^{(k)} \leftarrow P(z_t = k \mid Y_{t-h+1}, \ldots, Y_t)$

if $w_t^{(k)} \geq \tau$ **then**

$\mathcal{C}^{(k)} \leftarrow \mathcal{C}^{(k)} \cup \left\{\left(S_t,\, w_t^{(k)}\right)\right\}$

if $\left|\mathcal{C}^{(k)}\right| > N$ **then**

Drop oldest element from $\mathcal{C}^{(k)}$

end if

end if

end for

end for

Chapter 4

Coverage guarantees

A crucial property of conformal prediction is its ability to provide rigorous guarantees on predictive validity. Under the most general - ideally distribution-free - assumptions, can be shown that the proportion of prediction sets containing the true outcomes converges to $1 - \alpha$. Fundamental results in this direction were published in Gibbs and Candès, 2021 - where Propositions 1 and 2 were fully proven for ACP.

Assume that $\alpha_1 \in [0, 1]$ and that $\hat{Q}_t$ is non-decreasing with $\forall x < 0\ \hat{Q}_t(x) = -\infty$ and $\forall x > 1\ \hat{Q}_t(x) = +\infty$.

Proposition 1. *(Lemma 4.1 Gibbs and Candès, 2021)*
For all $t \in \mathbb{N}$

$$\alpha_t \in [-\gamma,\, 1 + \gamma].$$

Proposition 2. *(Proposition 4.1 Gibbs and Candès, 2021)*
For all $T \in \mathbb{N}$,

$$\left| \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t - \alpha \right| \leq \frac{\max\{\alpha_1,\, 1 - \alpha_1\} + \gamma}{T\gamma}.$$

Implies that

$$\lim_{T \to \infty} \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t = \alpha.$$

The proposed HMM-based error model maintains a separate parameter $\alpha_t^{(s)}$ for each hidden state updated according to the same recursive rule as in standard ACP. This structure allows inherit the coverage guarantees per state directly.

Proposition 3. *Let* $\{S_t\}_{t \in \mathbb{N}}$ *take values in a finite state set* $\mathcal{S}$ *with* $|\mathcal{S}| = N_S$*, and assume the weak-exogeneity condition* $S_t \perp \alpha_t^{(s)} \mid S_{t-1}$ *for all* $s \in \mathcal{S}$*. Then, for every realisation and every* $T \in \mathbb{N}$,

$$\left| \frac{1}{T} \sum_{t=1}^{T} \mathrm{err}_t - \alpha \right| \;\leq\; \frac{N_S \left(\max\{\alpha,\, 1 - \alpha\} + \gamma\right)}{\gamma\, T} \;\xrightarrow[T \to \infty]{}\; 0.$$

Proof. Let $T_s = |\{t \leq T : S_t = s\}|$. Applying Proposition 2 to this sub-sequence gives

$$\left| \sum_{t \,|\, S_t = s} \mathrm{err}_t - T_s\, \alpha \right| \;\leq\; \frac{\max\{\alpha,\, 1-\alpha\} + \gamma}{\gamma},$$

with the right side independent of T_s. Each step belongs to exactly one state, so $\sum_s T_s = T$ so with summing over s

$$\left| \sum_{t=1}^{T} \mathrm{err}_t - T\alpha \right| \;\leq\; \sum_{s \in \mathcal{S}} \frac{\max\{\alpha,\, 1-\alpha\} + \gamma}{\gamma} \;=\; \frac{N_S\,(\max\{\alpha,\, 1-\alpha\} + \gamma)}{\gamma}.$$

Dividing by T gives the stated bound. □

By Proposition 2 applied to each per-state sub-sequence, the same convergence also holds on every state visited infinitely often: $\frac{1}{T_s} \sum_{t \,|\, S_t = s} \mathrm{err}_t \to \alpha$. Coverage is therefore preserved not only globally but on every active state.

Chapter 5

Tests - Methodology and results

5.1 Methodology

All presented results are outcomes from a similar methodology applied across all tested methods each for {0.01, 0.05, 0.1} confidence interval.

5.1.1 Data Splitting

Each time series dataset is divided into four sequential parts:

- **Training set** – 4000 samples – used to train both the forecasting model and the Forecasting model.
- **Calibration set** – 2000 samples – used to compute nonconformity scores which are basis of quantile calibration and predictive interval intervals.
- **Validation set** – 4000 samples – used to evaluate the model on unseen observations and optimize methods hyperparameters.
- **Test set** – 4000 samples – used to evaluate the model on unseen observations using best-performing hyperparameters and providing final results.

5.1.2 Forecasting Model

Autoregresive predictions are generated using a Random Forest Regressor (Breiman, 2001). At each time step t, the model uses the four most recent observations as input features:

$$\hat{y}_t = RF(y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4}),$$

where $\hat{y}_t$ is the predicted value at time t.

5.1.3 Clustering model

State assignments are generated using a Hidden Markov Model (HMM). At each time step t, the model uses the previous h observations as input features and predicts the probabilities of all s states:

$$\{p_t^1, p_t^2, \ldots, p_t^s\} = \text{HMM}(v_{t-1}, v_{t-2}, \ldots, v_{t-h}),$$

where $\{p_t^i\}$ denotes the predicted probability of assigning sample to state i at time step t.

5.1.4 Conformal Prediction Methods

– **ACP** – Adaptive Conformal Prediction baseline, where the miscoverage level α_t is updated online at each time step according to:

$$\alpha_{t+1} = \alpha_t + \gamma \left(\alpha - \mathbf{1}\{y_t \notin \hat{C}_t\}\right),$$

where α is the target miscoverage level, γ is the step size, and $\hat{C}_t$ is the prediction interval at time t.

– **ACP + HMM (errors)** – extends ACP by using a Hidden Markov Model trained on prediction log-scaled errors of the forecasting model:

$$log(e_t) = log(|y_t - \hat{y}_t|),$$

$$\log(Error) \sim \mathcal{N}(\mu, \sigma^2) \implies Error \sim \text{Lognormal}(\mu, \sigma^2).$$

The errors emission distributions for each hidden state are assumed to be Gaussian. However, the observed variables are expressed on the logarithmic scale implies that the non-transformed observations follow a log-normal distribution. That operation is forced by fact that HMM frameworks support only modeling with Gaussian Mixtures.

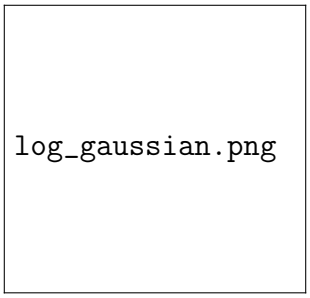


Figure 5.1: Log-normal distribution PDF
en.wikipedia.org/wiki/Log-normal_distribution

– **Ensemble batch prediction intervals (EnbPI)** – Xu and Xie, 2023a
trains B models on random subsamples of the training data. For each point - only those models that never saw it are averaged to form a prediction — making the resulting residuals unbiased without requiring a separate calibration set. These residuals create a sliding window, whose $1-\alpha$ percentile becomes the half-width of the prediction interval. As new observations arrive online, each residual enters the window and the oldest is dropped - like in the standard ACP.

– **Sequential Predictive Conformal Inference (SPCI)** – Xu and Xie, 2023b
similarly to EnbPI, trains a base model and collects residuals from the training phase. However, instead of taking a fixed empirical quantile from a sliding window it trains a quantile regression model Q on lagged residuals $(\varepsilon_{t-s}, \ldots, \varepsilon_{t-1})$ to predict the next quantile $q_t = Q(\varepsilon_{t-s}, \ldots, \varepsilon_{t-1}; 1-\alpha)$.

5.1.5 Hyperparameter Tuning

Hyperparameters for each method are optimized using Optuna with the TPE sampler, with the objective of minimizing the mean prediction interval width on the validation set. The choice of optimization technique is motivated by the characteristics of the ACP method. In its basic formulation, ACP includes a continuous hyperparameter γ which effectively determines the selection of a single discrete error value. This transition from a continuous parameter space to a discrete outcome introduces irregularities that are difficult to capture in techniques with better general convergence but using surrogate-models to approximate function behavior.

5.1.6 Evaluation Metrics

Each method is evaluated on the test set using two metrics:

– **Empirical coverage**:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbf{1}\{y_t \in \hat{C}_t\}.$$

A valid method is expected to achieve empirical coverage close to $1-\alpha$.

– **Mean interval width**:

$$\frac{1}{T}\sum_{t=1}^{T}\left(2\widehat{Q}_{1-\alpha_t}(S_{\mathrm{Cal}_t}\right).$$

A smaller width indicates in general a more efficient method.

– **Median interval width**:

$$\mathrm{median}_{t=1,\ldots,T}\left(2\widehat{Q}_{1-\alpha_t}(S_{\mathrm{Cal}_t})\right).$$

A smaller value indicates narrower typical intervals. The median is robust to outliers and is not overly influenced by a small number of extremely wide intervals.

5.1.7 Implementation details

The proofs presented previously assume the existence of infinite intervals. That makes metrics impossible to calculate and limits practical applicability. Usually to handle that problem we assume that lowest and highest error value in calibration set are considered as quantile value for respectively $\alpha < 0$ and $\alpha > 0$. To improve precision $\widehat{Q}_{1-\alpha}$ is calculated by interpolation. Also even if none of states has probability of assignment bigger than γ error will be added to most likely calibration set.

5.1.8 Plots

All presented plots in next sections contains results of experiments with setted calibration size 600.

5.2 Results - sythetic dataset

Dataset contains one dimensional observations of a sinusoidal signal jammed by noise - the noise variance switches between three regimes low, medium, and high volatility (according to a Markov chain).

$$\begin{pmatrix} 0.98 & 0.01 & 0.01 \\ 0.01 & 0.98 & 0.01 \\ 0.05 & 0.05 & 0.90 \end{pmatrix}$$

Table 5.1: Transition matrix for Markov chain generating regimes

synthetic_dataset.png

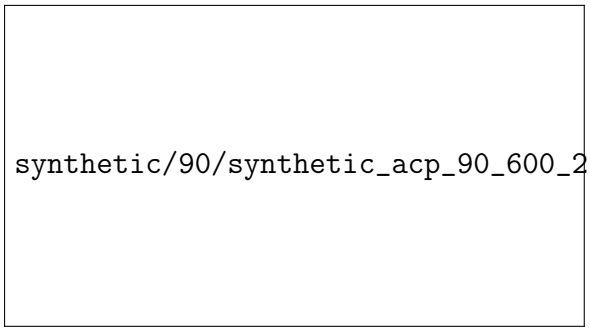
Figure 5.2: Synthetic dataset plot

5.2.1 Tests Summary

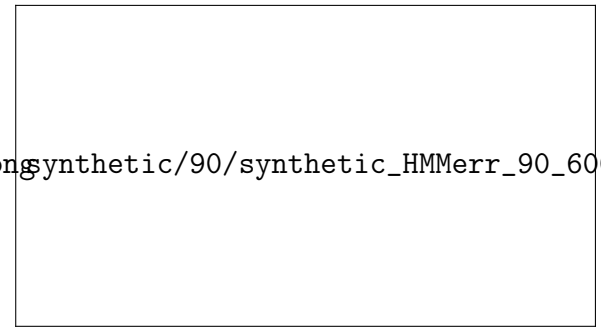
Cal size	Method	Coverage	Mean width	Median width
100	ACP	0.900	2.12	1.79
		0.950	3.04	2.39
		0.986	4.62	3.94
	ACP + HMM error	0.898	1.88 (-11.3%)	1.79 (0.0%)
		0.946	2.53 (-16.8%)	2.20 (-8.0%)
		0.983	4.33 (-6.3%)	3.73 (-5.3%)
300	ACP	0.899	1.92	1.72
		0.950	2.79	2.32
		0.988	5.45	5.19
	ACP + HMM error	0.895	1.76 (-8.3%)	1.58 (-8.1%)
		0.949	2.61 (-6.5%)	2.11 (-9.1%)
		0.991	4.28 (-21.5%)	3.14 (-39.5%)
600	ACP	0.899	1.90	1.70
		0.949	2.92	2.17
		0.990	5.25	4.87
	ACP + HMM error	0.891	1.69 (-11.1%)	1.46 (-14.1%)
		0.948	2.56 (-12.3%)	2.11 (-2.8%)
		0.989	3.98 (-24.2%)	3.14 (-35.5%)
	EnbPI	0.891	1.77 (-6.8%)	1.76 (+3.5%)
		0.945	2.43 (-16.8%)	2.43 (+12.0%)
		0.989	5.97 (+13.7%)	6.17 (+26.7%)
	SPCI	0.891	1.84 (-3.2%)	1.74 (+2.4%)
		0.942	2.35 (-19.5%)	2.28 (+5.1%)
		0.980	4.19 (-20.2%)	4.59 (-5.7%)

Table 5.2: Summary of results for the sythetic dataset across calibration set sizes. EnbPI and SPCI are trained on the full train+cal set (cal size = 600) and do not depend on calibration set size. Percentage difference compared to ACP. Coverage targets: 0.90, 0.95, 0.99.

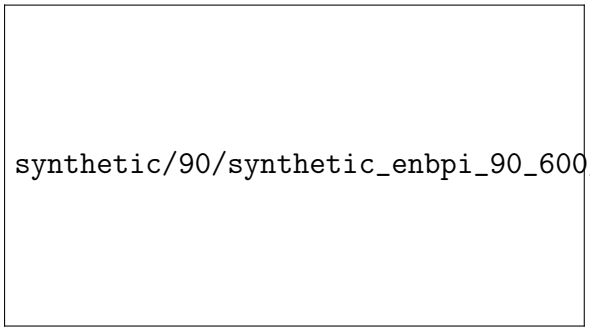
5.2.2 Evaluation plots - coverage 90%



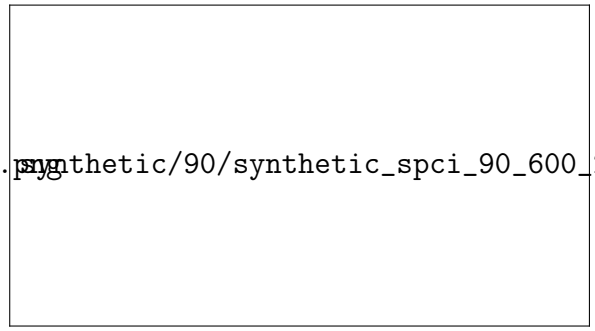
(a) ACP



(b) ACP + HMM error

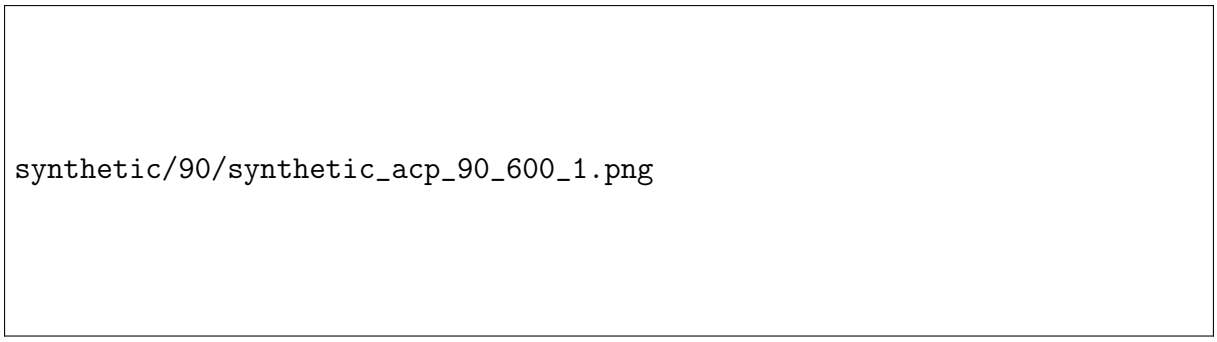


(c) EnbPI

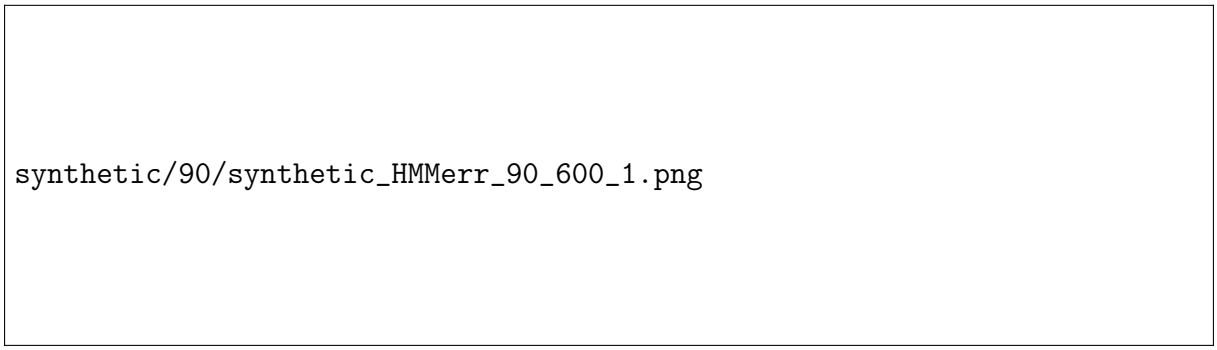


(d) SPCI

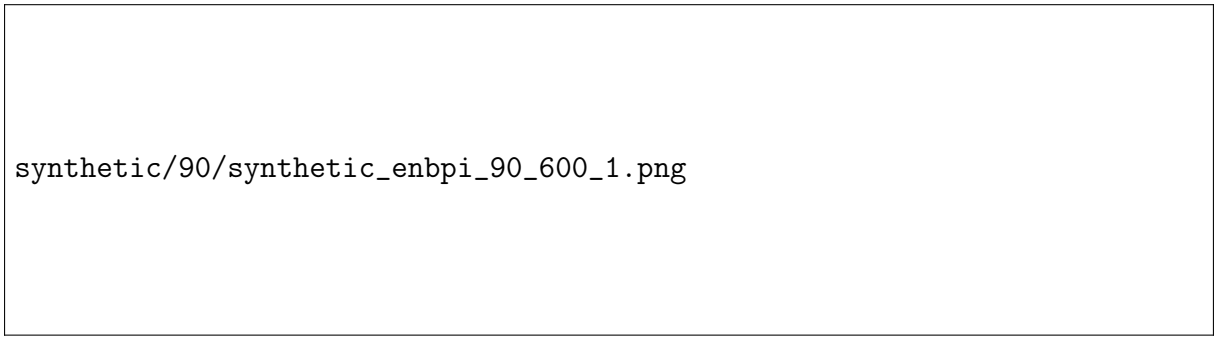
Figure 5.3: Interval width distribution for all methods at 90% coverage target.



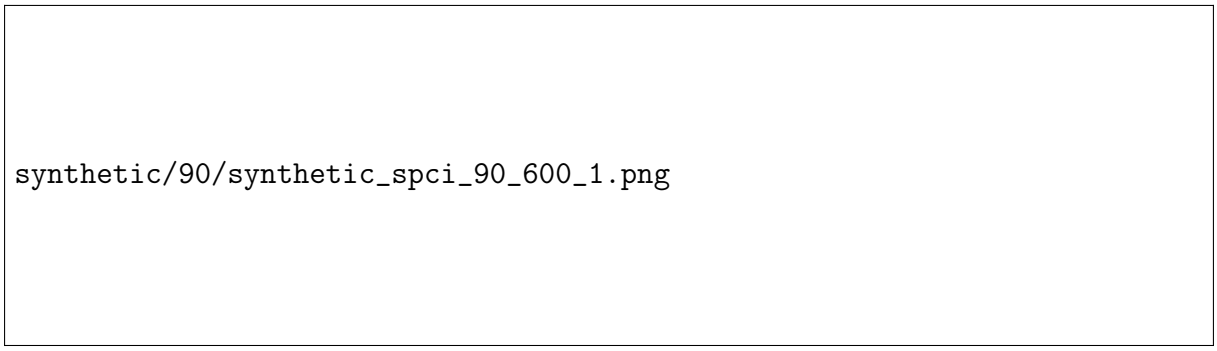
(a) ACP



(b) ACP + HMM error



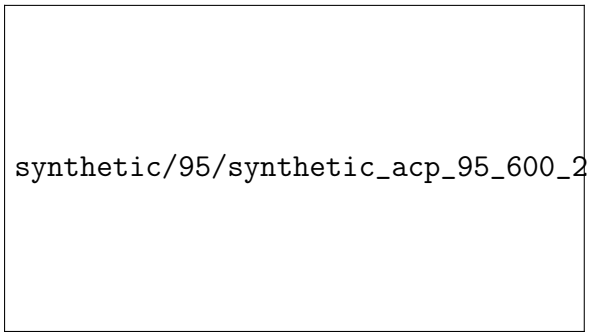
(c) EnbPI



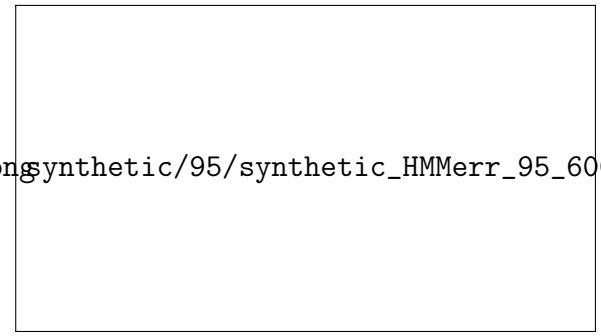
(d) SPCI

Figure 5.4: Interval widths changes for all methods at 90% coverage target

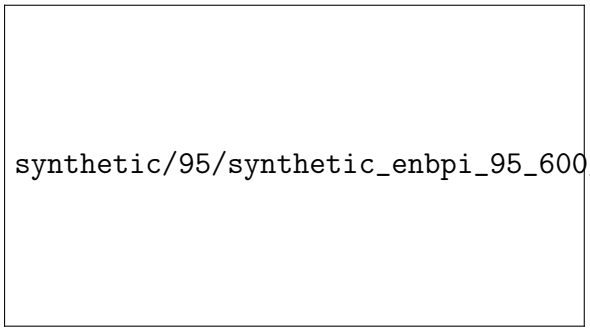
5.2.3 Evaluation plots - coverage 95%



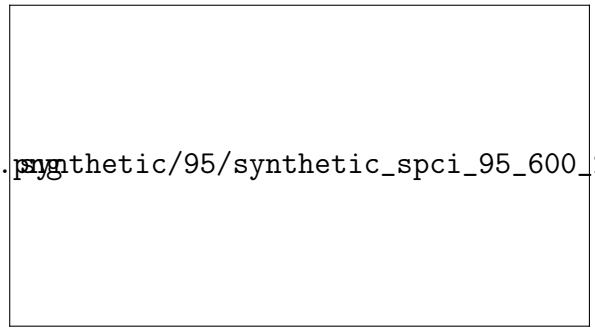
(a) ACP



(b) ACP + HMM error

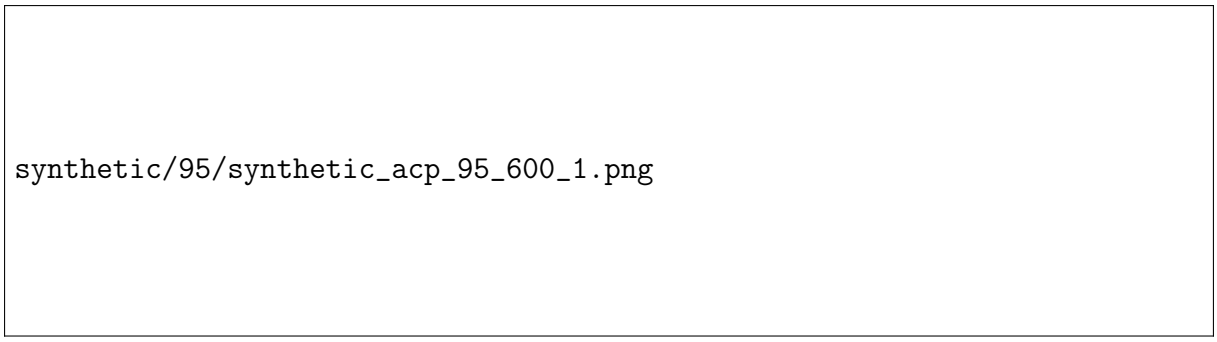


(c) EnbPI

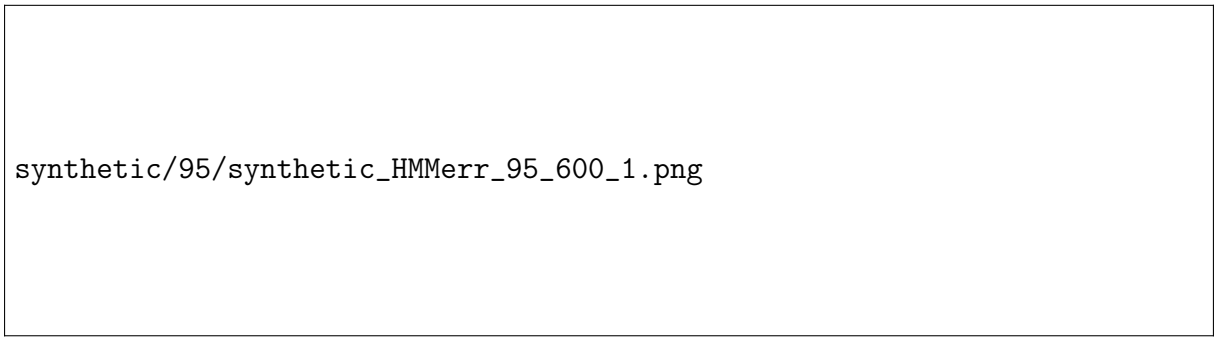


(d) SPCI

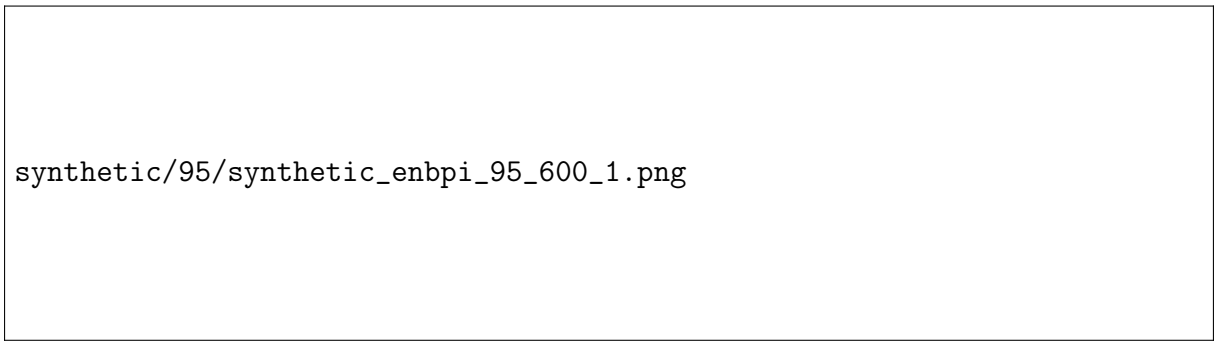
Figure 5.5: Interval width distribution for all methods at 95% coverage target.



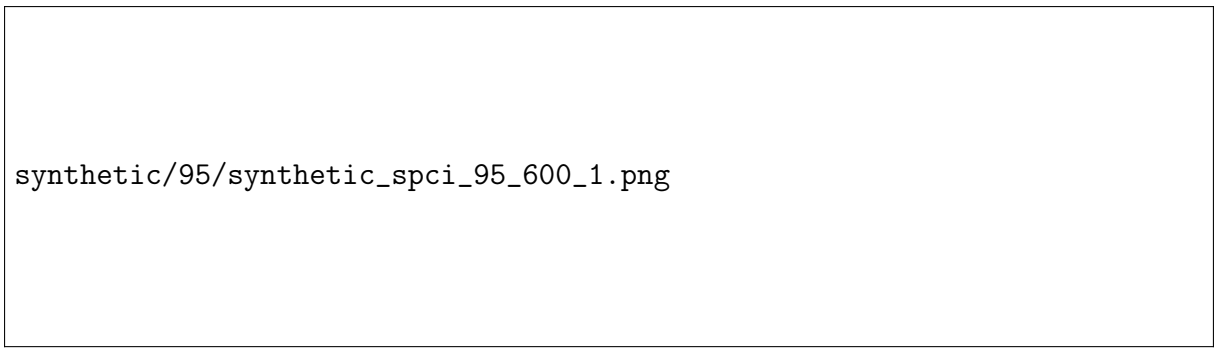
(a) ACP



(b) ACP + HMM error



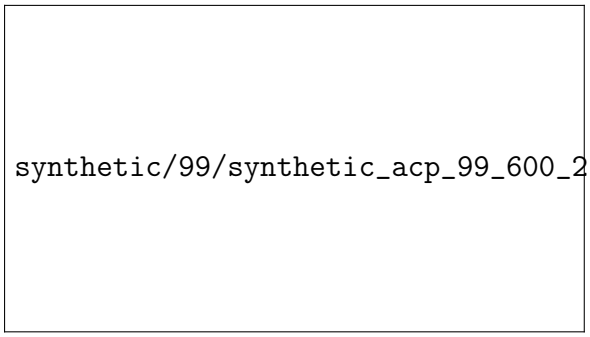
(c) EnbPI



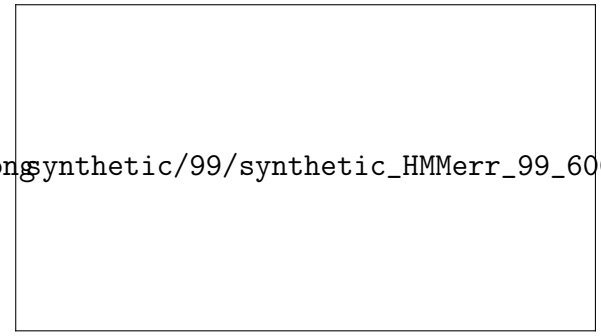
(d) SPCI

Figure 5.6: Interval widths changes for all methods at 95% coverage target

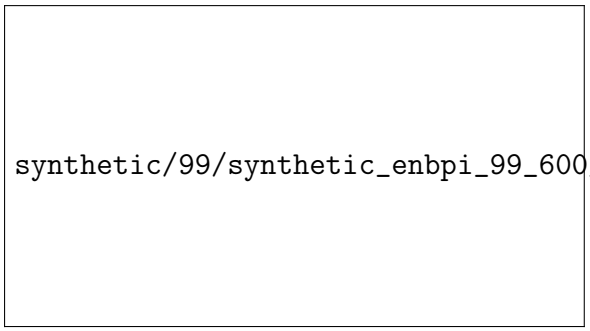
5.2.4 Evaluation plots - coverage 99%



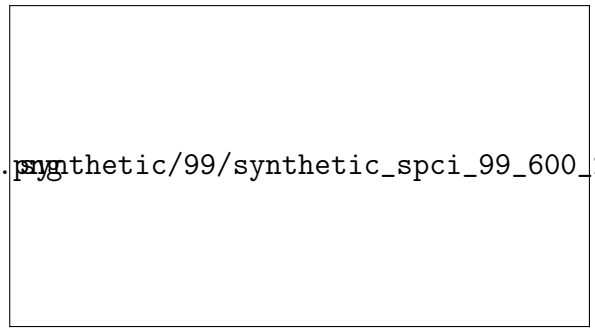
(a) ACP



(b) ACP + HMM error

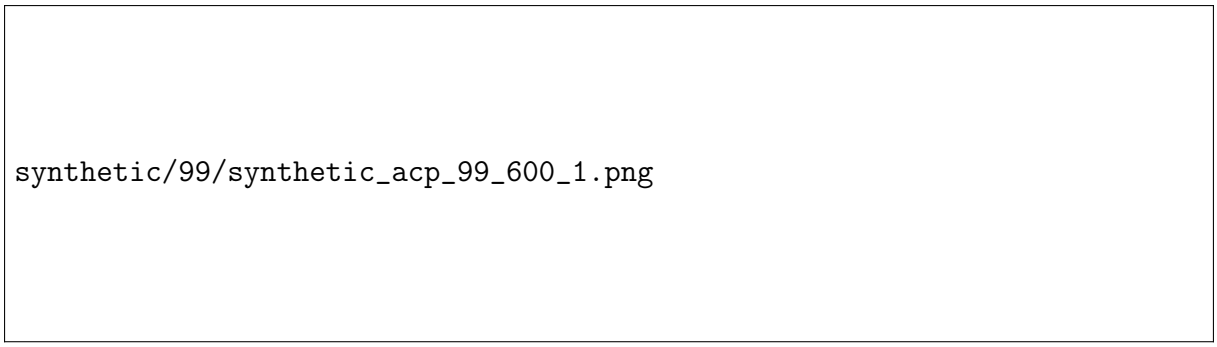


(c) EnbPI

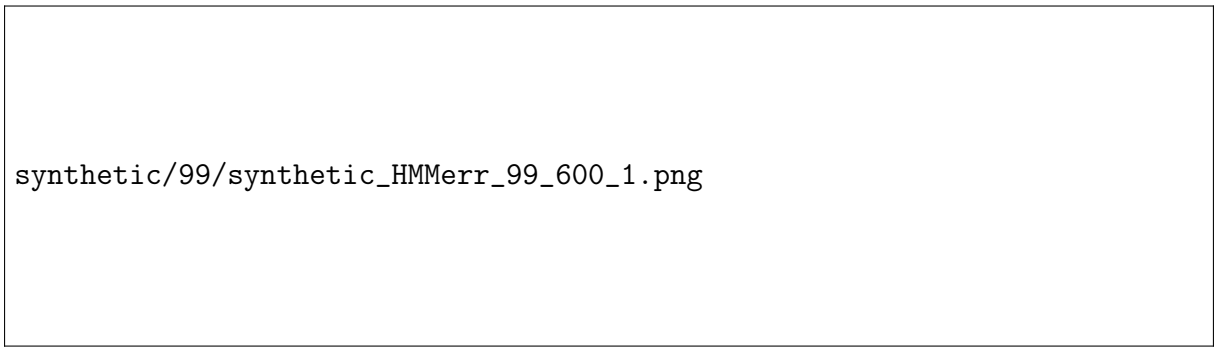


(d) SPCI

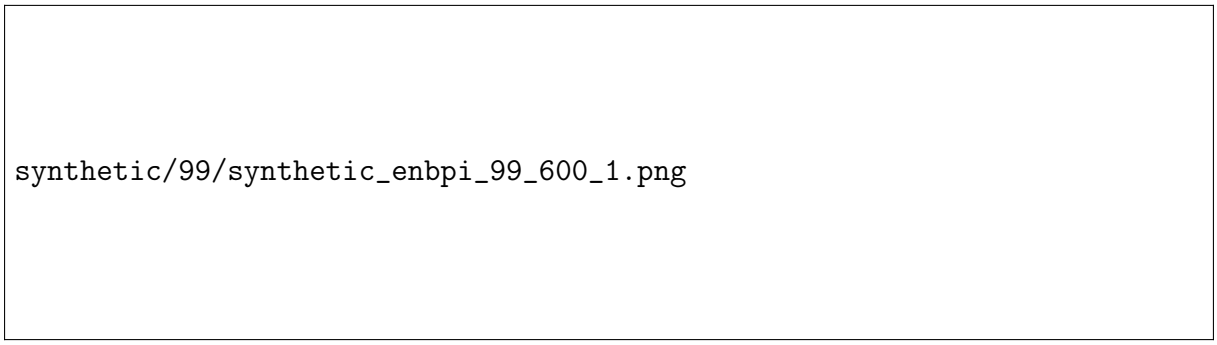
Figure 5.7: Interval width distribution for all methods at 99% coverage target.



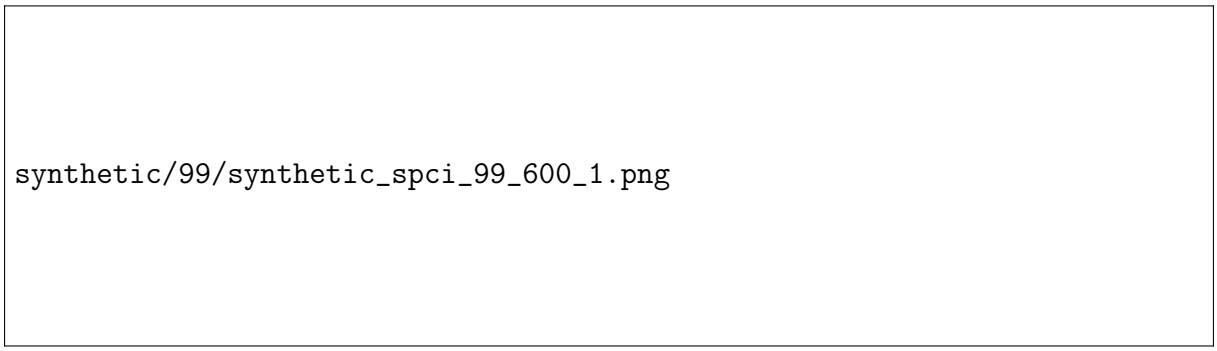
(a) ACP



(b) ACP + HMM error



(c) EnbPI



(d) SPCI

Figure 5.8: Interval widths changes for all methods at 99% coverage target

5.2.5 Tests conclusions

Coverage All evaluated methods achieve the target coverage with a maximum deviation of 0.01.

Interval Widths Increasing the calibration set size has a minimal positive effect on mean and median interval widths. Clustering-based methods consistently yield significant improvements in both metrics - reducing median width by up to 39.5% and mean width by up to 24.2% compared to standard ACP. SPCI achieves mean width reductions comparable to ACP + HMM error across most configurations. EnbPI performs similarly at lower coverage targets but deteriorates at the 0.99 level. Both EnbPI and SPCI produce notably wider median intervals than ACP, with increases of up to 26.7% and 5.7% respectively - in contrast to the consistent median reductions with clustering-based models.

Interval Width Over Time In clustering-based models, HMM state transitions lead to constant changes in interval widths, reflecting regime shifts present in the synthetic signal. In contrast, ACP plots has a progressive drift and EnbPI produces an almost constant width - unable to react to volatility changes. Intervals that fail to contain the true value are more readily identifiable in the ACP plots.

5.3 Results - electric power consumption dataset

Dataset contains one dimensional measurements of electric power usage from a household in France between 2006 and 2010. It includes variables such as active and reactive power, voltage, current intensity and energy usage. It is widely used as a baseline in forecasting and anomaly detection. Provided tests were conducted on power usage every single minute timeseries.



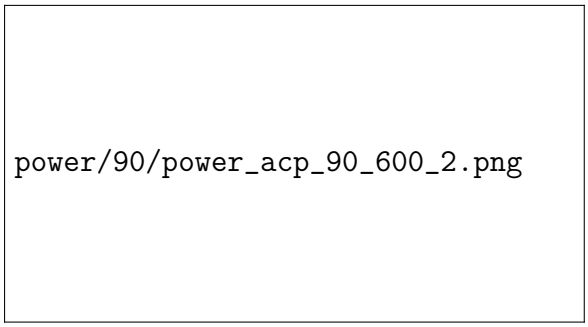
Figure 5.9: Power consumption dataset plot

5.3.1 Tests Summary

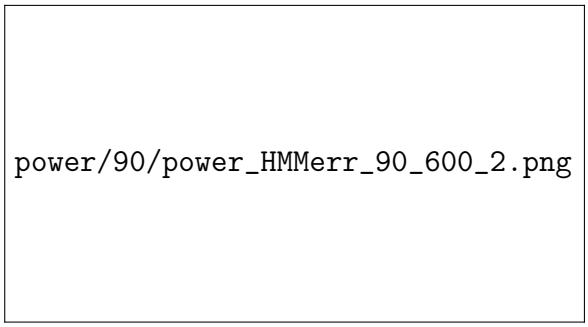
Cal size	Method	Coverage	Mean width	Median width
100	ACP	0.893	1.23	0.72
		0.949	1.73	1.14
		0.987	2.94	3.05
	ACP + HMM error	0.891	1.15 (−6.5%)	0.50 (−30.6%)
		0.941	1.54 (−11.0%)	0.84 (−26.3%)
		0.986	2.60 (−11.6%)	2.51 (−17.7%)
300	ACP	0.899	1.28	0.61
		0.948	1.86	1.14
		0.989	3.57	3.85
	ACP + HMM error	0.894	1.16 (−9.4%)	0.55 (−9.8%)
		0.945	1.66 (−10.7%)	0.87 (−23.7%)
		0.987	2.58 (−27.6%)	2.31 (−40.0%)
600	ACP	0.895	1.22	0.60
		0.948	1.81	1.01
		0.989	3.18	3.38
	ACP + HMM error	0.893	1.07 (−12.3%)	0.53 (−11.7%)
		0.948	1.58 (−12.7%)	0.83 (−17.8%)
		0.982	2.54 (−20.1%)	2.32 (−31.4%)
	EnbPI	0.869	0.82 (−32.8%)	0.77 (+28.3%)
		0.934	1.48 (−18.2%)	1.41 (+39.6%)
		0.985	3.77 (+18.6%)	3.59 (+6.2%)
	SPCI	0.887	0.83 (−32.0%)	0.68 (+13.3%)
		0.937	1.42 (−21.5%)	1.27 (+25.7%)
		0.982	2.71 (−14.8%)	2.73 (−19.2%)

Table 5.3: Summary of results for the power consumption dataset across calibration set sizes. EnbPI and SPCI are trained on the full train+cal set (cal size = 600) and do not depend on calibration set size. Percentage difference compared to ACP. Coverage targets: 0.90, 0.95, 0.99.

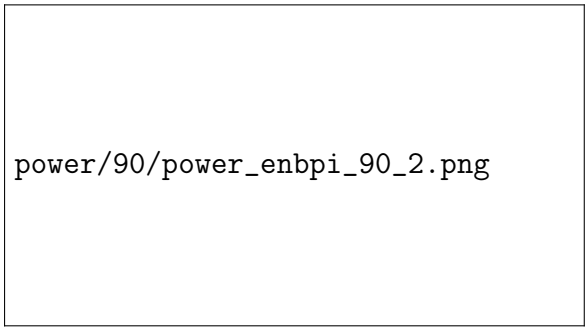
5.3.2 Evaluation plots - coverage 90%



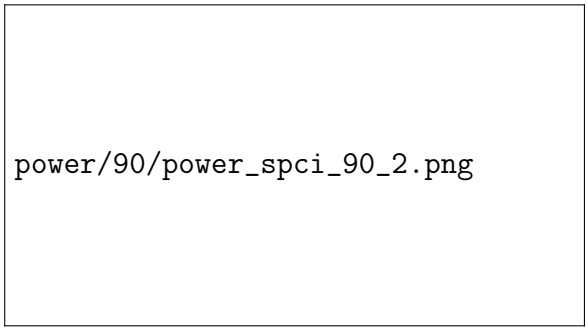
(a) ACP



(b) ACP + HMM error

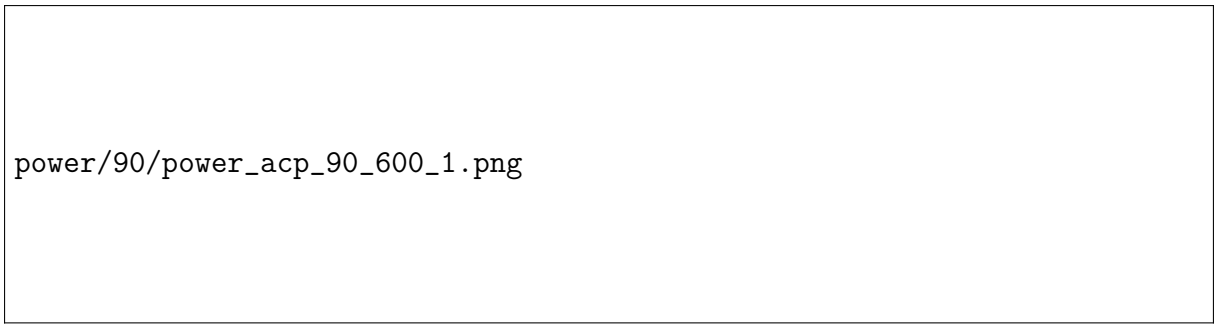


(c) EnbPI

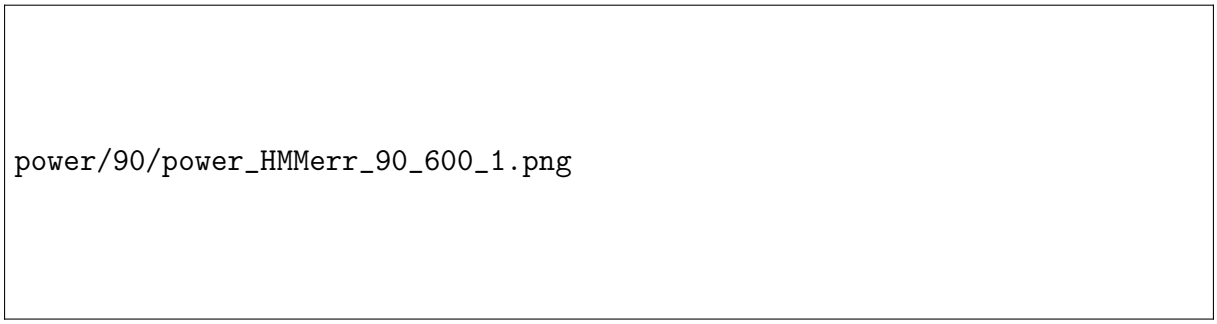


(d) SPCI

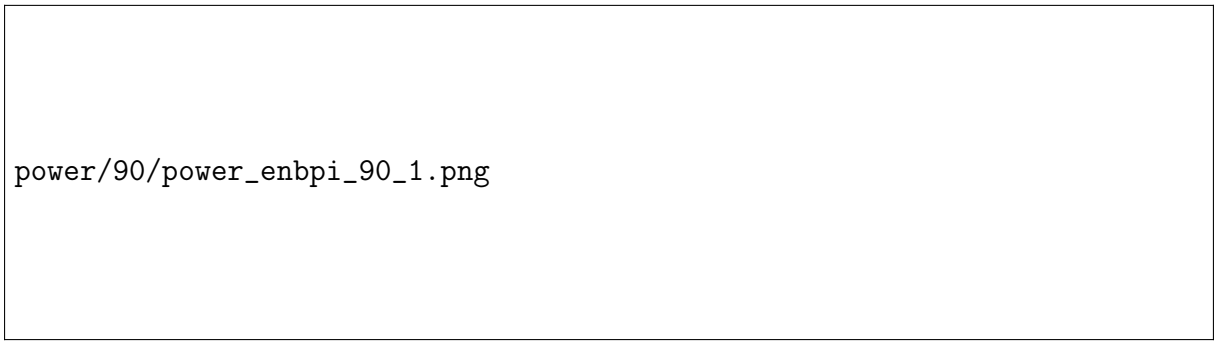
Figure 5.10: Interval width distribution for all methods at 90% coverage target.



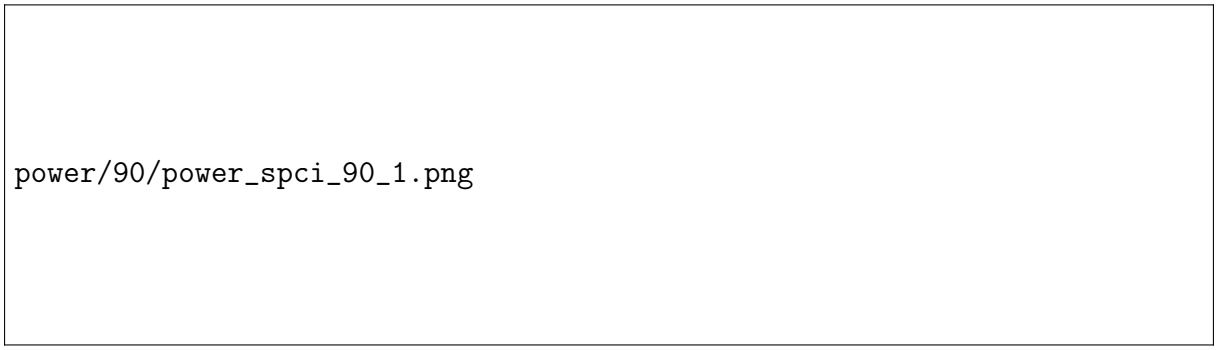
(a) ACP



(b) ACP + HMM error



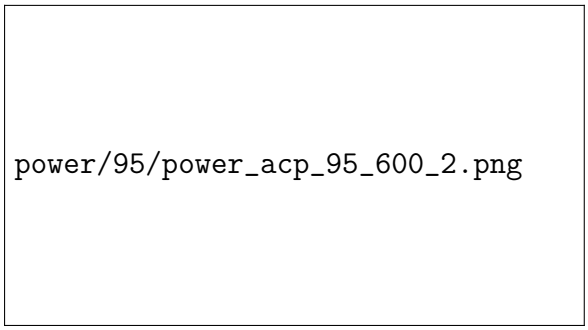
(c) EnbPI



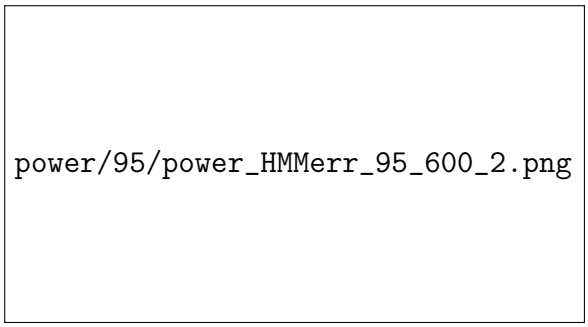
(d) SPCI

Figure 5.11: Interval widths changes for all methods at 90% coverage target

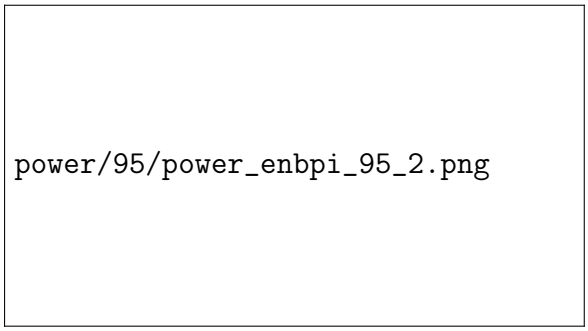
5.3.3 Evaluation plots - coverage 95%



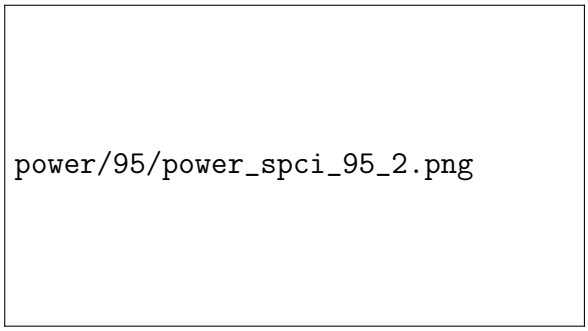
(a) ACP



(b) ACP + HMM error

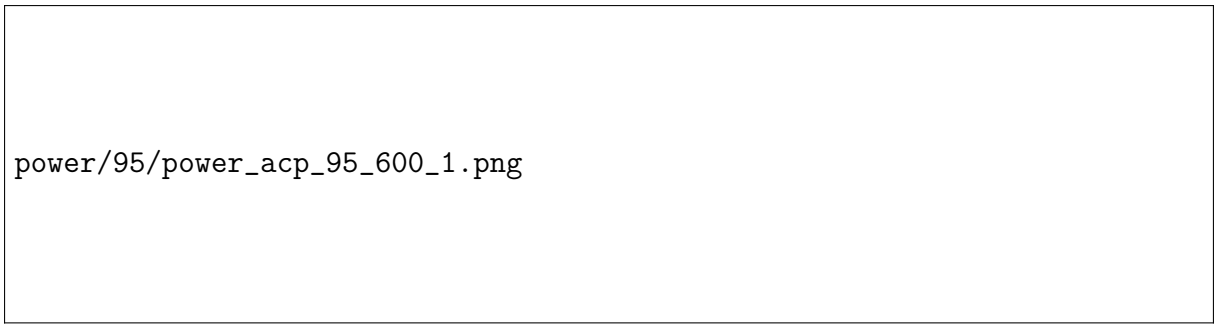


(c) EnbPI

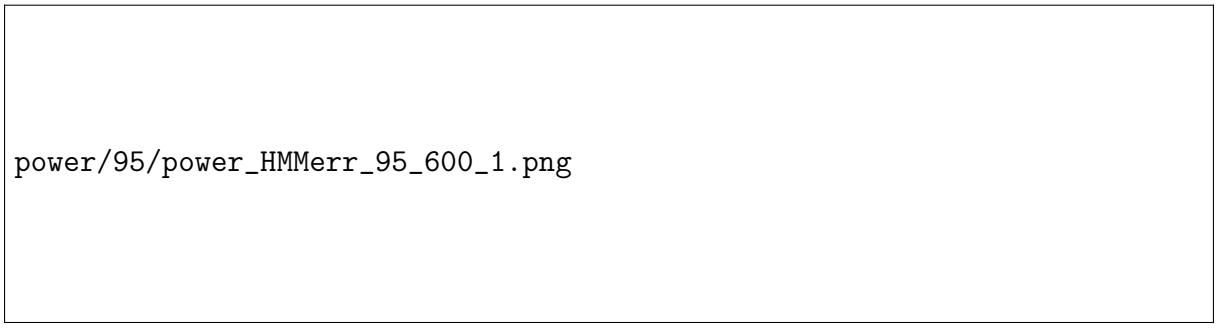


(d) SPCI

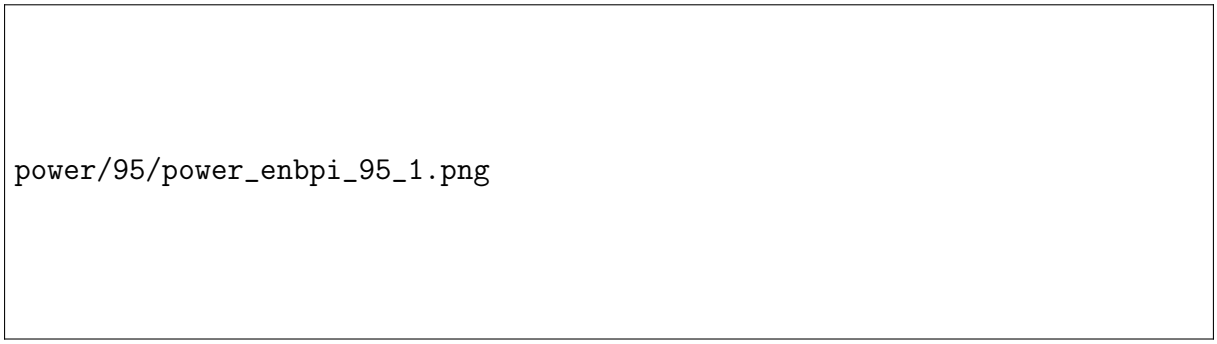
Figure 5.12: Interval width distribution for all methods at 95% coverage target



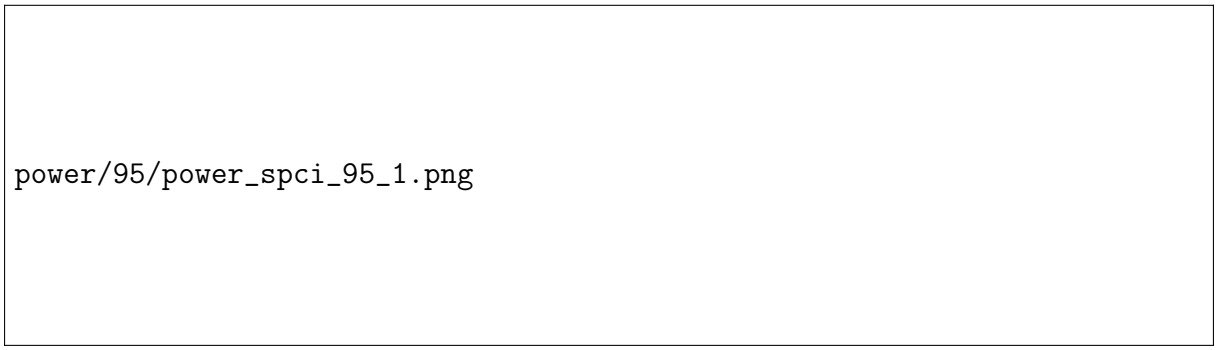
(a) ACP



(b) ACP + HMM error



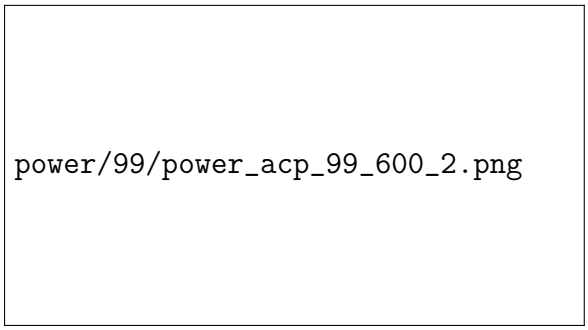
(c) EnbPI



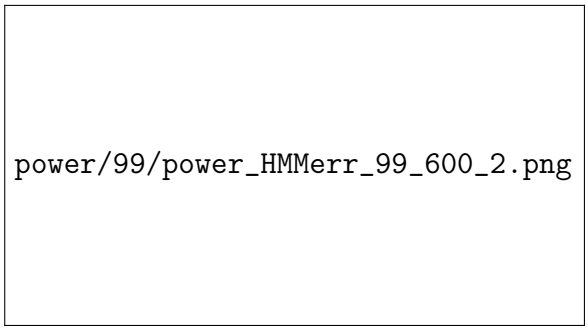
(d) SPCI

Figure 5.13: Interval widths changes for all methods at 95% coverage target

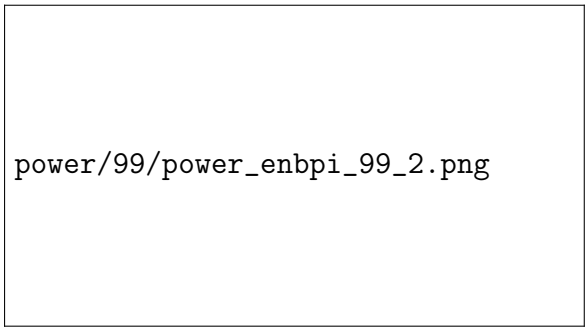
5.3.4 Evaluation plots - coverage 99%



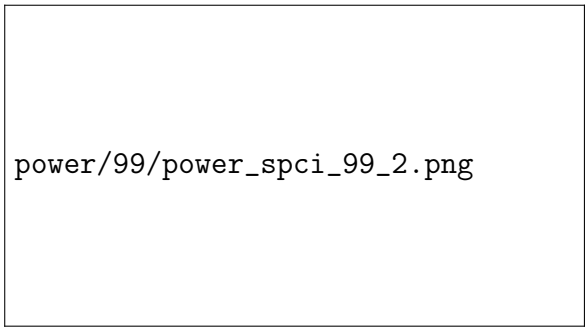
(a) ACP



(b) ACP + HMM error

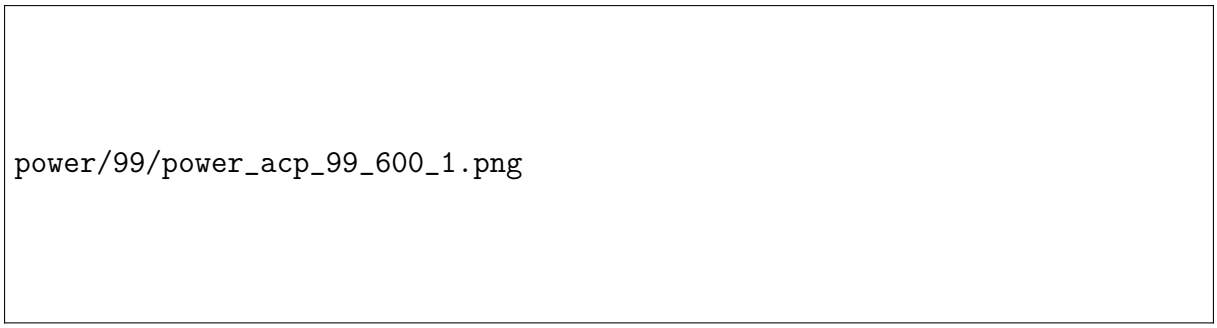


(c) EnbPI

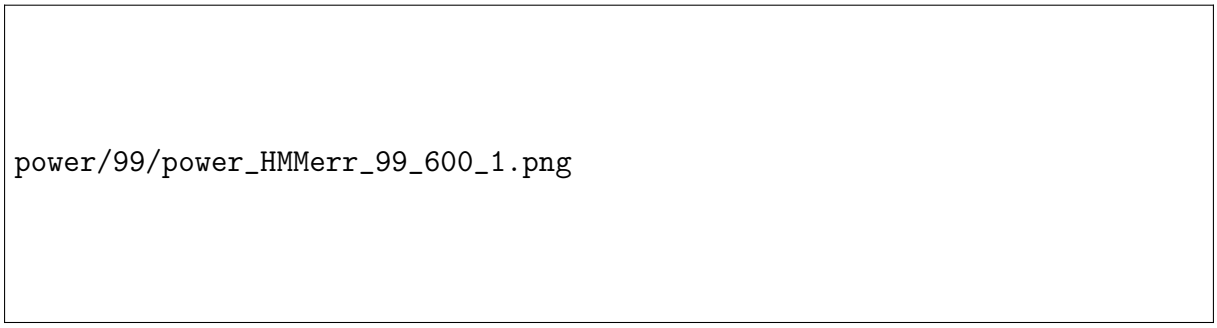


(d) SPCI

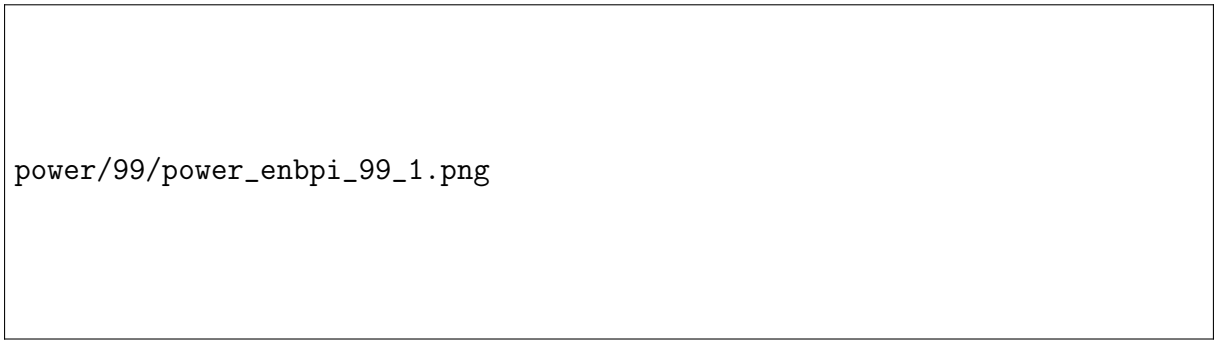
Figure 5.14: Interval width distribution for all methods at 99% coverage target



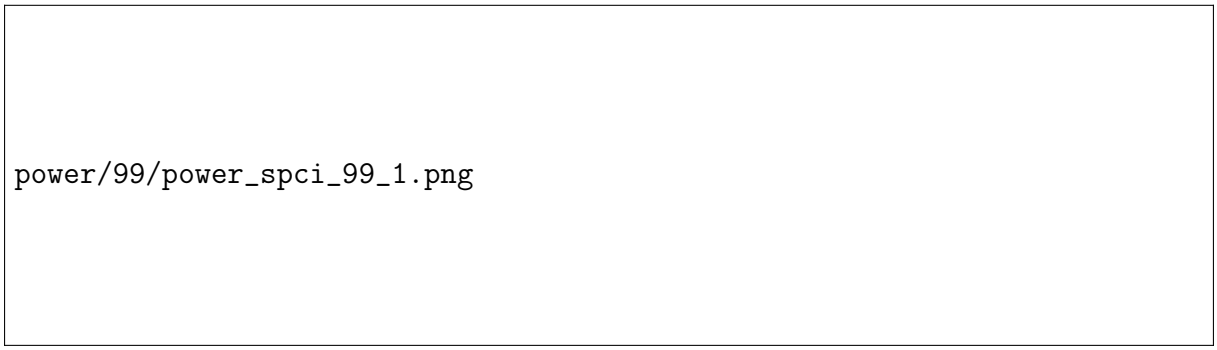
(a) ACP



(b) ACP + HMM error



(c) EnbPI



(d) SPCI

Figure 5.15: Interval widths changes for all methods at 99% coverage target

5.3.5 Tests conclusions

Coverage All evaluated methods achieve empirical coverage close to the nominal target. The largest deviation of 0.03 is observed for EnbPI at the 0.90 target level, while most

methods remain within 0.01-0.02 of the target across all coverage levels.

Interval Widths Increasing the calibration set size has a minimal positive effect on mean and median interval widths. Clustering-based methods consistently yield significant improvements in both metrics compared to standard ACP – reducing median width by up to 40.0% and mean width by up to 27.6%. SPCI achieves mean width reductions comparable to or exceeding ACP + HMM error across most configurations. EnbPI performs similarly at lower coverage targets, but deteriorates at the 0.99 level, where it produces wider intervals than ACP, making its mean width performance inconsistent overall. However, both EnbPI and SPCI produce notably wider median intervals than ACP across all tested configurations, with increases of up to 39.6% and 25.7% respectively - in contrast to the consistent median reductions delivered by clustering-based models.

Interval Width Over Time In clustering-based models HMM state transitions lead to abrupt changes in interval widths. EnbPI produces a smooth, slowly-varying width that adapts to long-term trends but fails to react to short-term regime changes. SPCI reacts more dynamically but still exhibits occasional sharp spikes.

5.4 Results - heartbeats dataset

Dataset contains one dimensional ECG recordings - commonly used for analyzing and detecting cardiac arrhythmias. Provided tests were conducted on one-dimensional timeseries - MLII potential every single. MLII is a configuration that records the electrical potential difference along the axis from the right arm to the left leg. It is one of the most widely used datasets in biomedical signal processing and machine learning research.

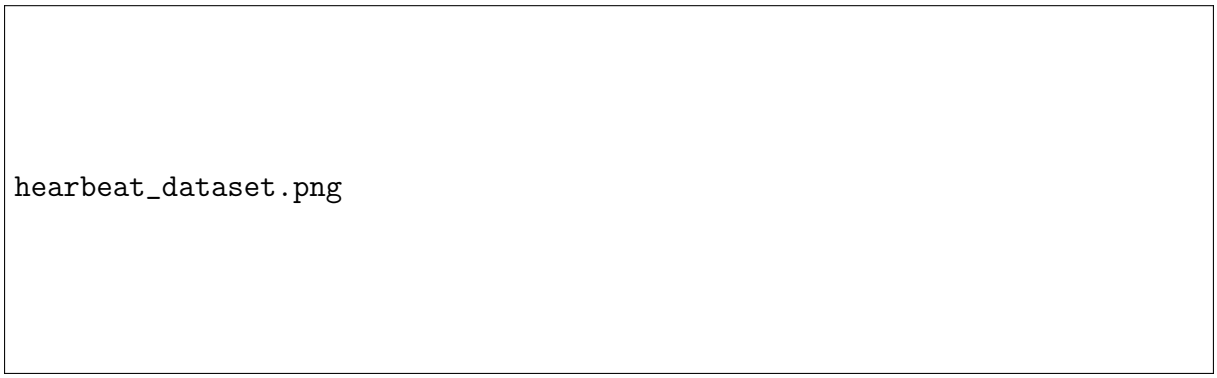


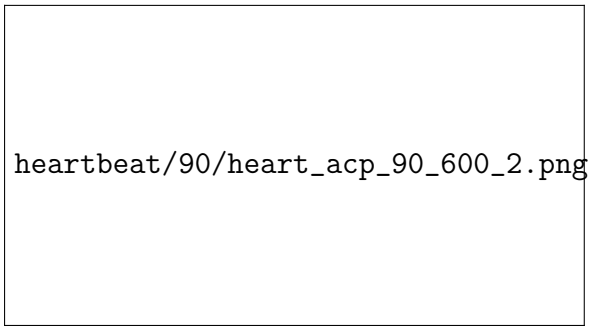
Figure 5.16: Heartbeats dataset plot

5.4.1 Tests Summary

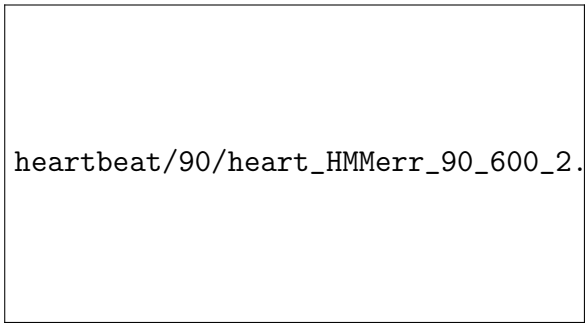
Cal size	Method	Coverage	Mean width	Median width
100	ACP	0.903	24.97	20.52
		0.953	38.68	32.07
		0.980	61.35	48.72
	ACP + HMM error	0.905	23.76 (−4.8%)	15.29 (−25.5%)
		0.952	33.19 (−14.2%)	19.33 (−39.7%)
		0.986	52.13 (−15.0%)	37.07 (−23.9%)
300	ACP	0.902	23.765	19.29
		0.951	34.825	27.79
		0.991	96.071	91.24
	ACP + HMM error	0.903	21.669 (−8.8%)	14.92 (−22.7%)
		0.955	32.373 (−7.0%)	18.40 (−33.8%)
		0.991	63.317 (−34.1%)	26.23 (−71.3%)
600	ACP	0.900	21.62	18.16
		0.950	31.55	26.00
		0.991	74.50	74.80
	ACP + HMM error	0.902	21.69 (+0.3%)	14.56 (−19.8%)
		0.954	31.20 (−1.1%)	18.34 (−29.5%)
		0.992	57.78 (−22.4%)	28.22 (−62.3%)
	EnbPI	0.866	17.78 (−17.8%)	17.94 (−1.2%)
		0.936	27.31 (−13.4%)	27.81 (+7.0%)
		0.987	67.52 (−9.4%)	67.88 (−9.3%)
	SPCI	0.787	14.36 (−33.6%)	14.34 (−21.0%)
		0.848	19.52 (−38.1%)	19.77 (−24.0%)
		0.887	34.12 (−54.2%)	34.70 (−53.6%)

Table 5.4: Summary of results for the heartbeats dataset across calibration set sizes. EnbPI and SPCI are trained on the full train and calibration set (cal size = 600) and do not depend on calibration set size. Percentage difference compared to ACP. Coverage targets: 0.90, 0.95, 0.99.

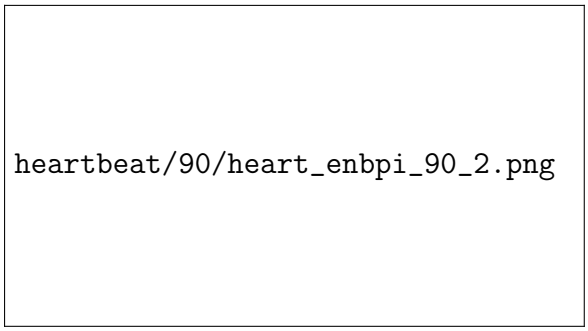
5.4.2 Evaluation plots - coverage 90%



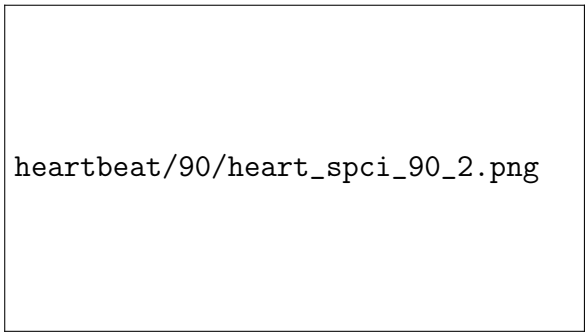
(a) ACP



(b) ACP + HMM error

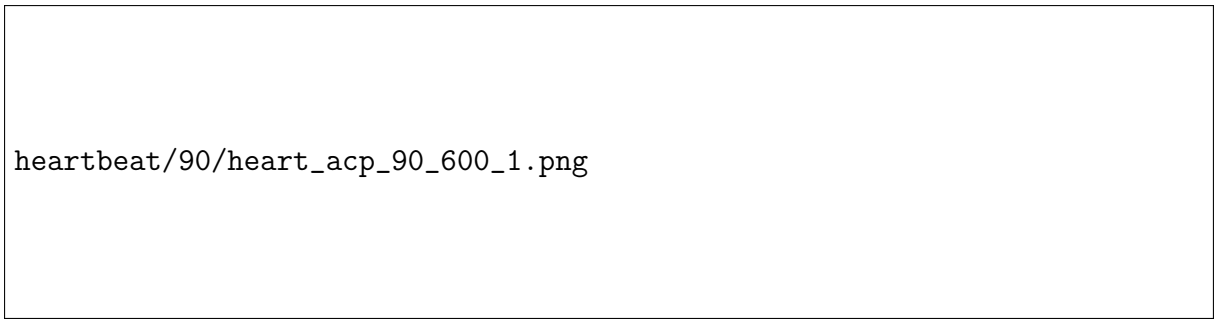


(c) EnbPI

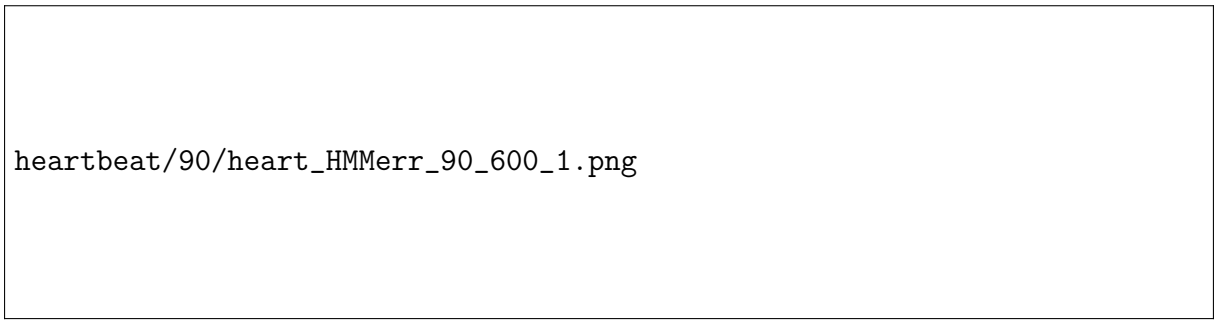


(d) SPCI

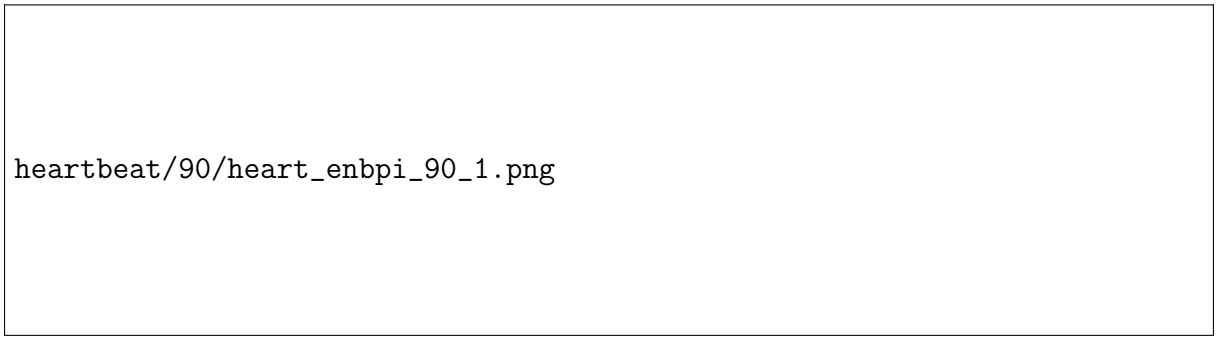
Figure 5.17: Interval width distribution for all methods at 90% coverage target.



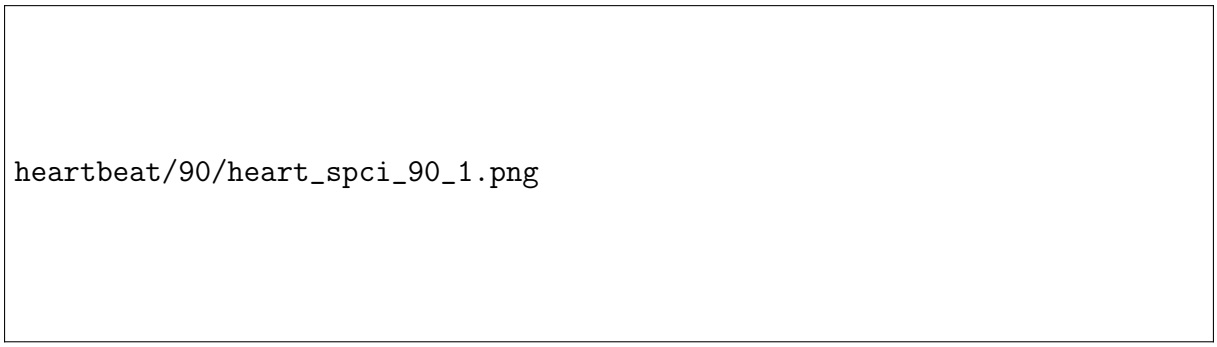
(a) ACP



(b) ACP + HMM error



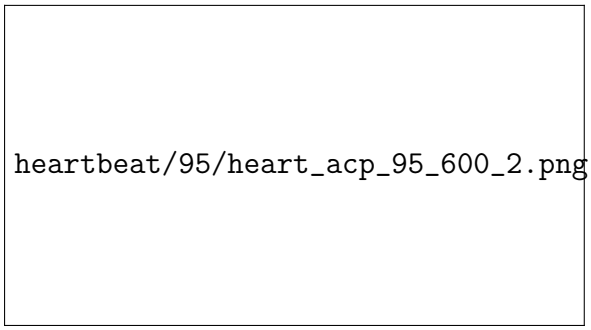
(c) EnbPI



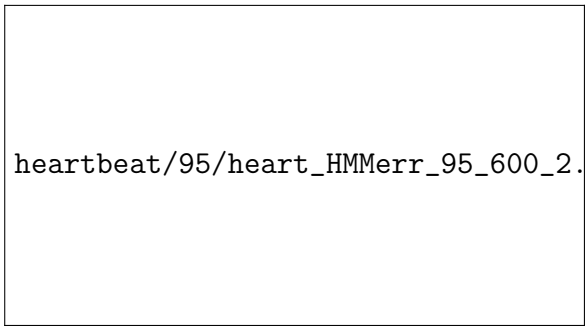
(d) SPCI

Figure 5.18: Interval widths changes for all methods at 90% coverage target

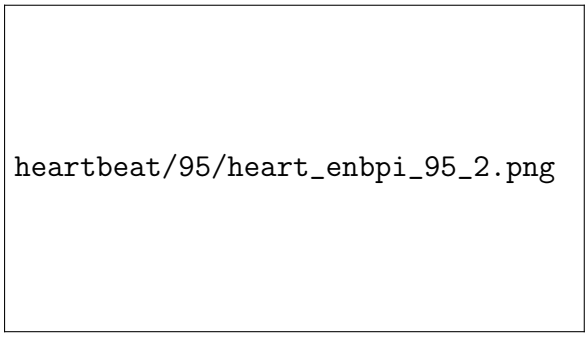
5.4.3 Evaluation plots - coverage 95%



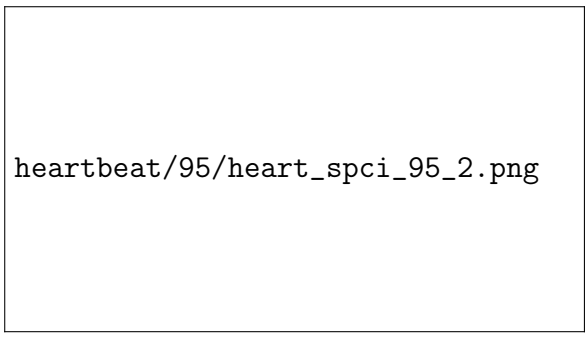
(a) ACP



(b) ACP + HMM error

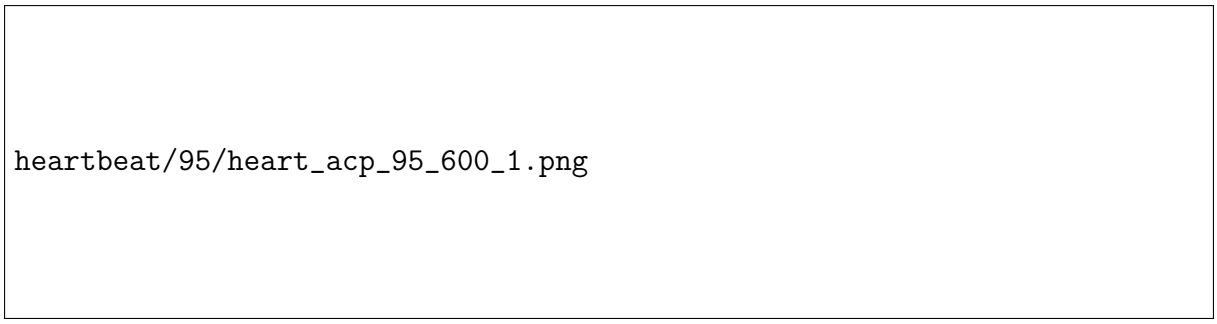


(c) EnbPI

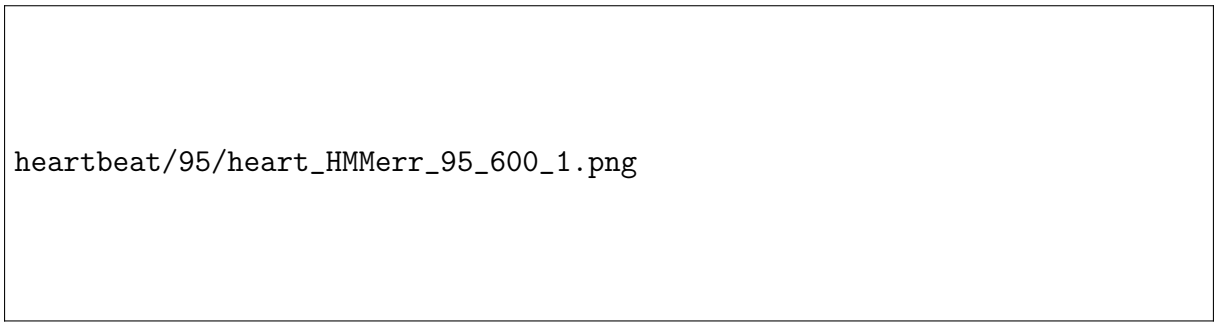


(d) SPCI

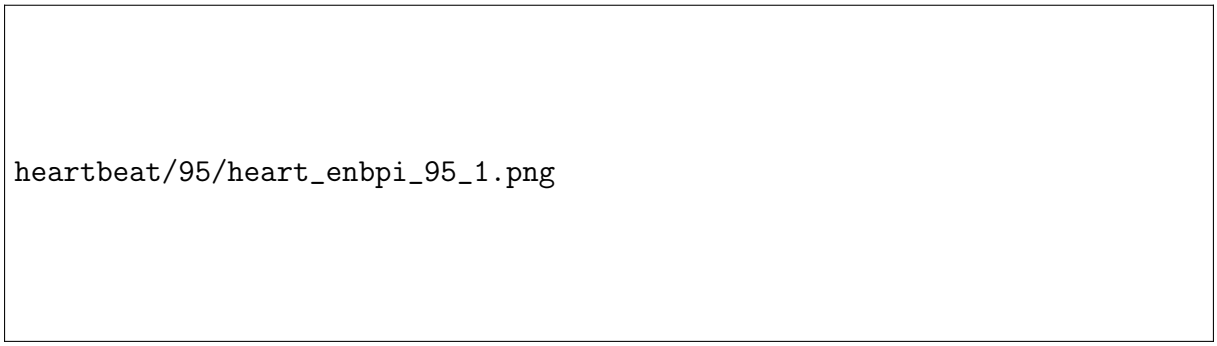
Figure 5.19: Interval width distribution for all methods at 95% coverage target.



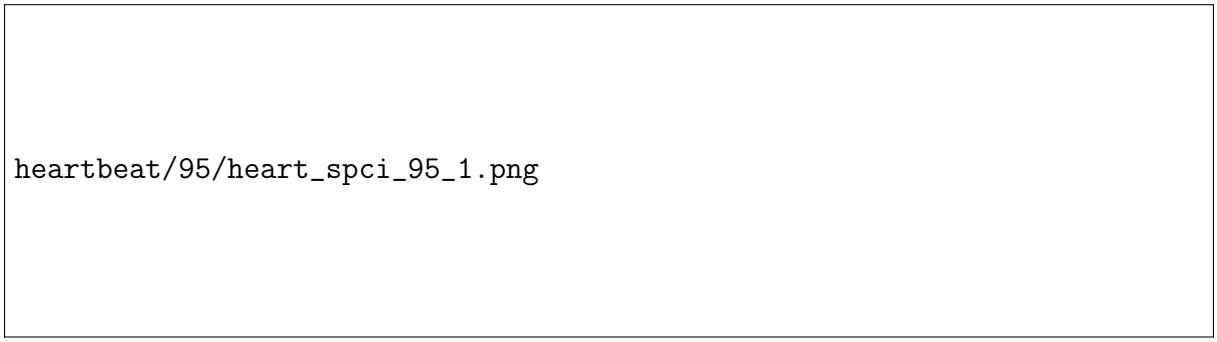
(a) ACP



(b) ACP + HMM error



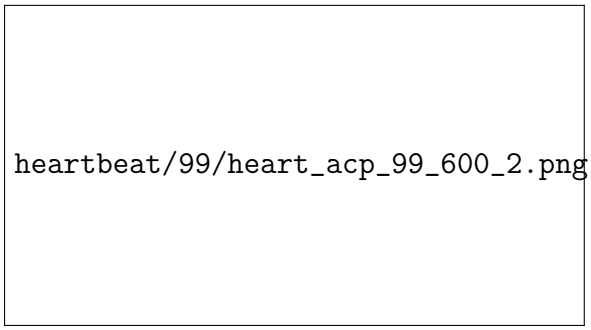
(c) EnbPI



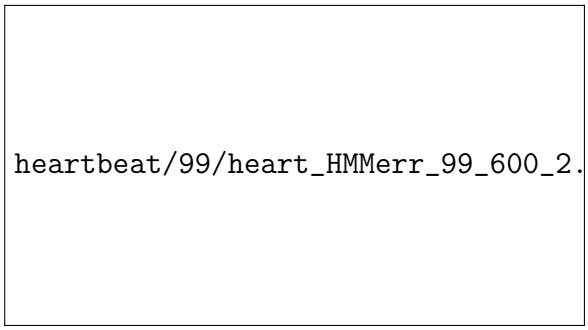
(d) SPCI

Figure 5.20: Interval widths changes for all methods at 95% coverage target

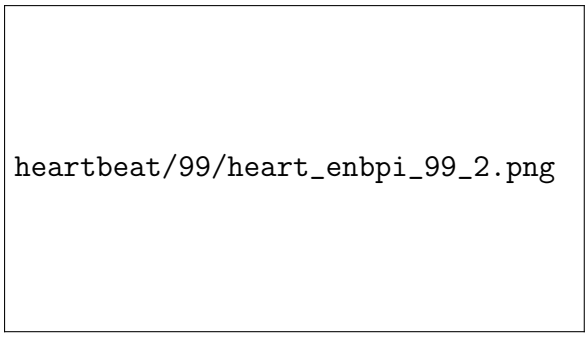
5.4.4 Evaluation plots - coverage 99%



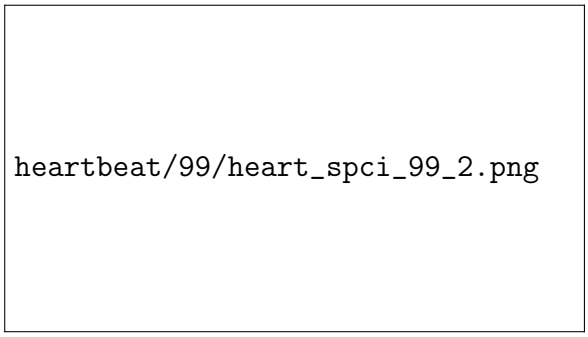
(a) ACP



(b) ACP + HMM error

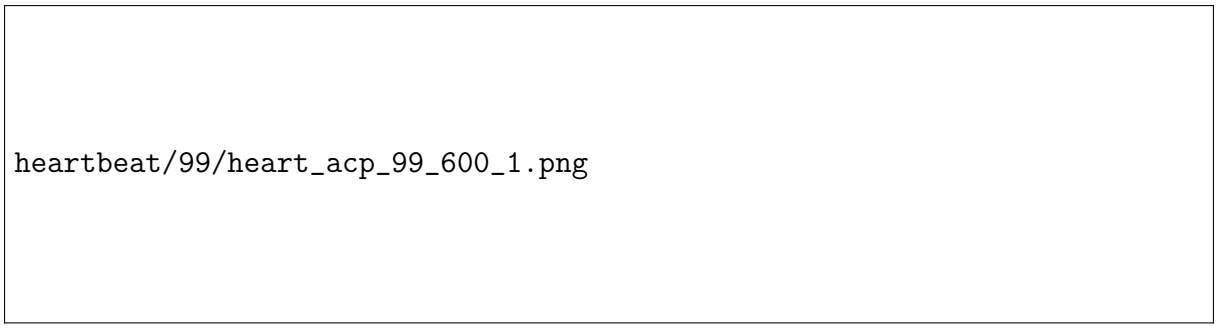


(c) EnbPI

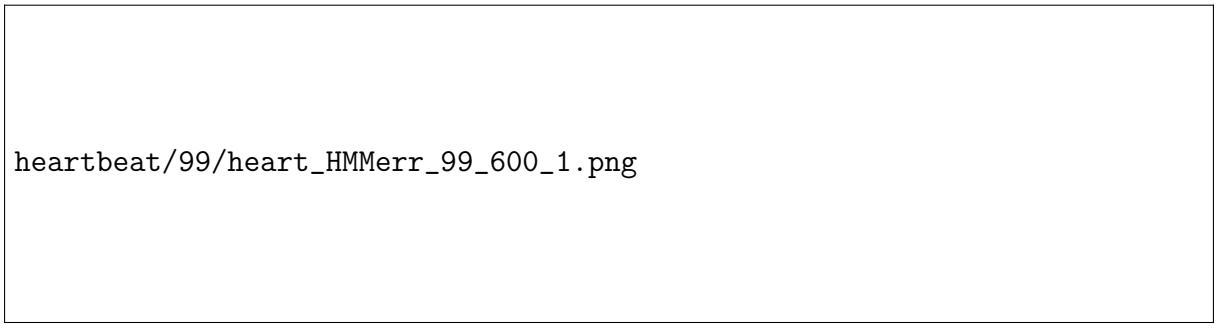


(d) SPCI

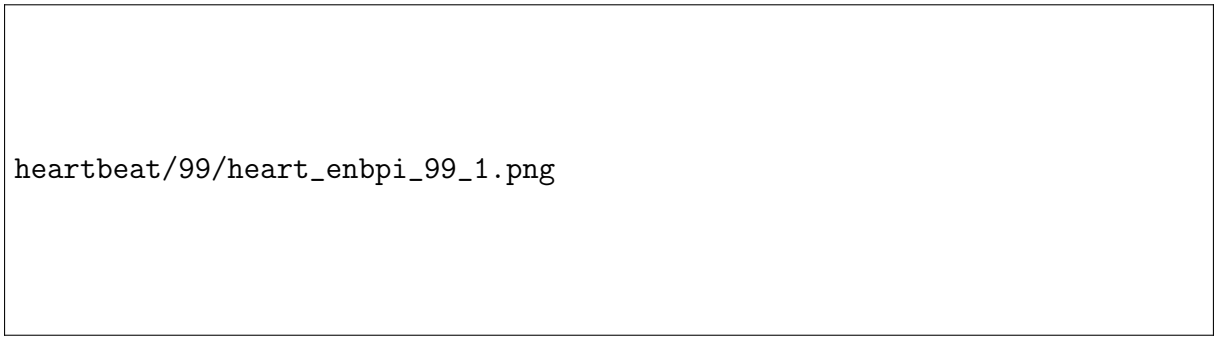
Figure 5.21: Interval width distribution for all methods at 99% coverage target.



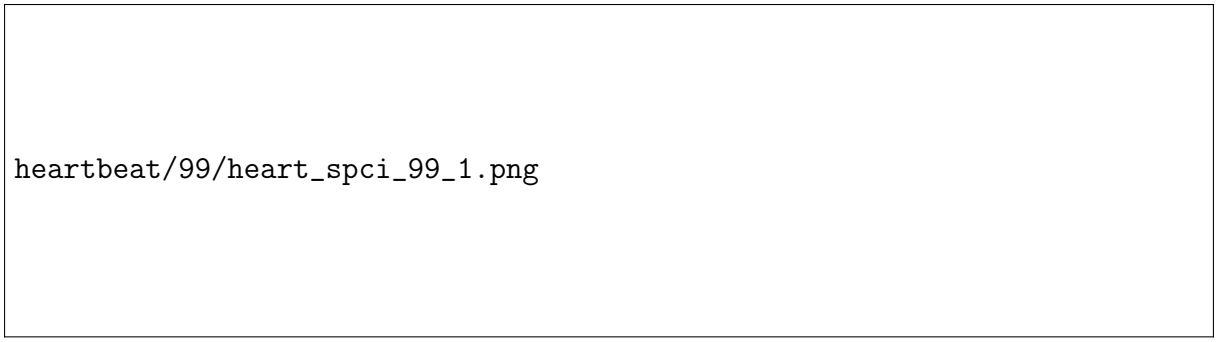
(a) ACP



(b) ACP + HMM error



(c) EnbPI



(d) SPCI

Figure 5.22: Interval widths changes for all methods at 99% coverage target

5.4.5 Tests conclusions

Coverage Coverage results on the heartbeats dataset reveal an important limitation of SPCI - empirical coverage falls significantly below the target across all levels, reaching as

low as 0.787 at the 0.90 target. EnbPI also undercovers at the 0.90 level with a deviation of 0.034. Clustering-based methods and standard ACP maintain coverage within 0.01 of the target.

Interval Widths Clustering-based methods again yield consistent improvements in median width - by up to 71.3% - while mean width not exceeding 34.1% with easy to see grouped intervals widths on histogram. EnbPI achieves mean width reductions of up to 17.8%, with median width largely comparable to ACP. SPCI produces the largest mean width reductions of up to 54.2%, but these results must be interpreted with a view of significant coverage violations.

Interval Width Over Time The clustering-based model exhibits characteristic switching between narrow and wide intervals driven by HMM state transitions. EnbPI produces a smoothly increasing width over time, suggesting it adapts to global drift but is insensitive to local regime changes. SPCI shows highly variable widths with frequent sharp spikes.

5.5 Execution time

method	**time [s]**
ACP	3.853
ACP + HMM	7.945
SPCI	341.555
EnbPI	7.190

Table 5.5: Example of training and prediction combined execution time. (MacBook Pro, M3 Pro 11C CPU, 14C GPU 18/512GB SSD, dataset size 10000).

5.6 Optimal hyperparameters

Since the clustering-based approach divided prediction errors into groups with similar characteristics and weighted quantile may result easier jumping between values with high weights, the optimal values of parameter γ were significantly lower than in the standard ACP method and were typically close to the imposed lower bound 0.001.

Dataset	$(\mathbf{1}-\alpha)$	$\lvert C\rvert$	γ
Synthetic	0.90	100	0.089
Synthetic	0.95	100	0.093
Synthetic	0.99	100	0.199
Synthetic	0.90	300	0.040
Synthetic	0.95	300	0.017
Synthetic	0.99	300	0.003
Synthetic	0.90	600	0.038
Synthetic	0.95	600	0.060
Synthetic	0.99	600	0.019
Hearbeats	0.90	100	0.001
Hearbeats	0.95	100	0.001
Hearbeats	0.99	100	0.200
Hearbeats	0.90	300	0.001
Hearbeats	0.95	300	0.008
Hearbeats	0.99	300	0.001
Hearbeats	0.90	600	0.009
Hearbeats	0.95	600	0.009
Hearbeats	0.99	600	0.001
Electric power	0.90	100	0.001
Electric power	0.95	100	0.008
Electric power	0.99	100	0.199
Electric power	0.90	300	0.036
Electric power	0.95	300	0.038
Electric power	0.99	300	0.106
Electric power	0.90	600	0.016
Electric power	0.95	600	0.033
Electric power	0.99	600	0.023

Table 5.6: Optimal hyperparameters for ACP method

Dataset	$(\mathbf{1}-\alpha)$	$\lvert C\rvert$	γ	states number	τ	HMM window
Synthetic	0.90	100	0.001	3	0.375	9
Synthetic	0.95	100	0.001	4	0.307	8
Synthetic	0.99	100	0.001	4	0.351	8
Synthetic	0.90	300	0.001	3	0.399	8
Synthetic	0.95	300	0.020	4	0.171	6
Synthetic	0.99	300	0.001	4	0.351	8
Synthetic	0.90	600	0.001	3	0.364	8
Synthetic	0.95	600	0.011	3	0.203	8
Synthetic	0.99	600	0.001	4	0.385	6
Hearbeats	0.90	100	0.001	4	0.385	6
Hearbeats	0.95	100	0.001	4	0.322	7
Hearbeats	0.99	100	0.001	4	0.150	7
Hearbeats	0.90	300	0.016	4	0.255	9
Hearbeats	0.95	300	0.001	4	0.413	6
Hearbeats	0.99	300	0.002	4	0.485	6
Hearbeats	0.90	600	0.105	4	0.247	5
Hearbeats	0.95	600	0.002	4	0.368	6
Hearbeats	0.99	600	0.001	4	0.206	5
Electric power	0.90	100	0.001	4	0.371	6
Electric power	0.95	100	0.001	4	0.173	6
Electric power	0.99	100	0.001	3	0.332	7
Electric power	0.90	300	0.001	4	0.360	8
Electric power	0.95	300	0.007	4	0.400	6
Electric power	0.99	300	0.001	3	0.465	9
Electric power	0.90	600	0.001	4	0.171	5
Electric power	0.95	600	0.001	3	0.260	6
Electric power	0.99	600	0.004	3	0.322	7

Table 5.7: Optimal hyperparameters for clustering-based ACP method

Chapter 6

Summary

This thesis addresses the problem of uncertainty quantification in time series forecasting using conformal prediction. The goal of this work was to extend the Adaptive Conformal Prediction framework to handle regime-switching behavior by integrating a clustering model into the calibration pipeline and preserving formal coverage guaranties.

The main contribution is a clustering-based ACP algorithm that uses a Hidden Markov Model to identify regimes in the time series. Instead of a single shared calibration set the method separate observations and switches between them. Groups are updated online with weighted nonconformity scores reflecting the posterior state probability. On the theoretical side Proposition 3 proves that the proposed method inherits the asymptotic coverage guarantee of ACP - the miscoverage error converges to the target level α, both globally and on every state.

The method was evaluated on three datasets - synthetic sinusoidal signal, electric power consumption and ECG heartbeat recordings - with 3 coverage targets - 0.90, 0.95 and 0.99. The clustering-based method consistently achieved the target coverage while producing substantially narrower prediction intervals than standard ACP, with median width reductions of up to 71.3% and mean width reductions of up to 34.1%. Compared to EnbPI and SPCI, the proposed method offered more reliable coverage and at least comparable metrics reductions. SPCI violated mostly at the coverage guaranty on the heartbeat data set (empirical coverage as low as 0.787 with a 0.90 target) and EnbPI failed at the 0.90 target with 0.034 deviation.

Several directions remain open for future work. The most promising extension concerns the adaptive selection of the parameter γ value. In the current formulation γ is fixed as a hyperparameter, but allowing it to vary dynamically - for example by increasing γ when consecutive miscoverage events are detected and decreasing it during stable periods. Further directions could also focused on clustering improvement with online clusters reparameterization (ex. Cappé, 2009).

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Appendix A

Glossary of Notations

Symbol	Description
α	Target miscoverage level (expected probability that the true value is not in the predicted interval).
$1-\alpha$	Target coverage level or confidence level.
D_{Tr}	Training set, used to train the forecasting model.
D_{Cal}	Calibration set, used to compute nonconformity scores.
D_{Te}	Test set, used to evaluate model performance.
$\hat{\mu}(\cdot)$	Forecasting model (e.g., Random Forest).
$Y_t, \hat{Y}_t$	True value and predicted value of the time series at time t, respectively.
C, C_t	Set of nonconformity scores (residuals) used for calibration.
$\lvert Y_i - \hat{\mu}(X_i)\rvert$	Nonconformity score, typically defined as the absolute residual.
n	Size of the calibration set.
$\hat{Q}_{1-\alpha}$	Empirical $(1-\alpha)$-quantile of the nonconformity scores.
$\hat{C}_\alpha(X_t)$	Conformal prediction interval generated for a given observation.
γ	Step-size parameter used for the recursive update of the miscoverage level α_t.
α_t	Effective miscoverage level updated at each time step.
err_t	Error indicator variable (1 if Y_t is not covered by the interval $\hat{C}_{\alpha_t}(X_t)$).
S	Size of the sliding window for the calibration set in standard ACP.

Table A.1 Notations used in thesis

Symbol	Description
h	Sequence length (window) of previous observations used as input for the clustering model.
τ	Probability threshold for updating state-specific calibration sets.
$z_t, \hat{z}_t$	Hidden state (cluster) at time t and its estimated value.
$C(k)_t$	State-specific conformal calibration set for state k.
$w_t^{(k)}$	Posterior probability of the system being in state k at time t.
$\tilde{w}_i$	Normalized weights used to calculate the weighted empirical quantile.